2547160_Lab1_2

June 11, 2026

1 Lab 01 : Data preprocessing and visualisation

1.1 Task 1 : Understanding Data

[1]:
```
import pandas as pd
import numpy as np
```

[2]:
```
city= pd.read_csv('./data/city_day.csv')
city.head()
```

[2]:

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	AQI_Bucket
0	Ahmedabad	2015-01-01	NaN	NaN	0.92	18.22	17.15	NaN	0.92	27.64	133.36	0.00	0.02	0.00	NaN	NaN
1	Ahmedabad	2015-01-02	NaN	NaN	0.97	15.69	16.46	NaN	0.97	24.55	34.06	3.68	5.50	3.77	NaN	NaN
2	Ahmedabad	2015-01-03	NaN	NaN	17.40	19.30	29.70	NaN	17.40	29.07	30.70	6.80	16.40	2.25	NaN	NaN
3	Ahmedabad	2015-01-04	NaN	NaN	1.70	18.48	17.97	NaN	1.70	18.59	36.08	4.43	10.14	1.00	NaN	NaN
4	Ahmedabad	2015-01-05	NaN	NaN	22.10	21.42	37.76	NaN	22.10	39.33	39.31	7.01	18.89	2.78	NaN	NaN

[3]:
```
crop=pd.read_csv('./data/crop_production.csv')
crop.head()
```

[3]:

	State_Name	District_Name	Crop_Year	Season	Crop	Area	Production
0	Andaman and Nicobar Islands	NICOBARS	2000	Kharif	Arecanut	1254.0	2000.0
1	Andaman and Nicobar Islands	NICOBARS	2000	Kharif	Other Kharif pulses	2.0	1.0
2	Andaman and Nicobar Islands	NICOBARS	2000	Whole Year	Rice	102.0	321.0
3	Andaman and Nicobar Islands	NICOBARS	2000	Whole Year			
4	Andaman and Nicobar Islands	NICOBARS	2000	Whole Year			

```
3              Banana   176.0       641.0
4           Cashewnut   720.0       165.0
```

1.1.1 City Dataset

Structure and Shape

[4]:
```
city.shape
```

[4]:
```
(29531, 16)
```

[5]:
```
city.dtypes
```

[5]:
```
City           object
Date           object
PM2.5         float64
PM10          float64
NO            float64
NO2           float64
NOx           float64
NH3           float64
CO            float64
SO2           float64
O3            float64
Benzene       float64
Toluene       float64
Xylene        float64
AQI           float64
AQI_Bucket     object
dtype: object
```

[6]:
```
city.columns
```

[6]:
```
Index(['City', 'Date', 'PM2.5', 'PM10', 'NO', 'NO2', 'NOx', 'NH3', 'CO', 'SO2',
       'O3', 'Benzene', 'Toluene', 'Xylene', 'AQI', 'AQI_Bucket'],
      dtype='object')
```

Summary Statistics

[7]:
```
city.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29531 entries, 0 to 29530
Data columns (total 16 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   City        29531 non-null  object
 1   Date        29531 non-null  object
 2   PM2.5       24933 non-null  float64
 3   PM10        18391 non-null  float64
```

```
 4   NO           25949 non-null  float64
 5   NO2          25946 non-null  float64
 6   NOx          25346 non-null  float64
 7   NH3          19203 non-null  float64
 8   CO           27472 non-null  float64
 9   SO2          25677 non-null  float64
 10  O3           25509 non-null  float64
 11  Benzene      23908 non-null  float64
 12  Toluene      21490 non-null  float64
 13  Xylene       11422 non-null  float64
 14  AQI          24850 non-null  float64
 15  AQI_Bucket   24850 non-null  object
dtypes: float64(13), object(3)
memory usage: 3.6+ MB
```

[8]:
```
city.describe()
```

[8]:

	PM2.5	PM10	NO	NO2	NOx	\
count	24933.000000	18391.000000	25949.000000	25946.000000	25346.000000	
mean	67.450578	118.127103	17.574730	28.560659	32.309123	
std	64.661449	90.605110	22.785846	24.474746	31.646011	
min	0.040000	0.010000	0.020000	0.010000	0.000000	
25%	28.820000	56.255000	5.630000	11.750000	12.820000	
50%	48.570000	95.680000	9.890000	21.690000	23.520000	
75%	80.590000	149.745000	19.950000	37.620000	40.127500	
max	949.990000	1000.000000	390.680000	362.210000	467.630000	

	NH3	CO	SO2	O3	Benzene	\
count	19203.000000	27472.000000	25677.000000	25509.000000	23908.000000	
mean	23.483476	2.248598	14.531977	34.491430	3.280840	
std	25.684275	6.962884	18.133775	21.694928	15.811136	
min	0.010000	0.000000	0.010000	0.010000	0.000000	
25%	8.580000	0.510000	5.670000	18.860000	0.120000	
50%	15.850000	0.890000	9.160000	30.840000	1.070000	
75%	30.020000	1.450000	15.220000	45.570000	3.080000	
max	352.890000	175.810000	193.860000	257.730000	455.030000	

	Toluene	Xylene	AQI
count	21490.000000	11422.000000	24850.000000
mean	8.700972	3.070128	166.463581
std	19.969164	6.323247	140.696585
min	0.000000	0.000000	13.000000
25%	0.600000	0.140000	81.000000
50%	2.970000	0.980000	118.000000
75%	9.150000	3.350000	208.000000
max	454.850000	170.370000	2049.000000

[9]:
```
city.describe(include='object')
```

[9]:

	City	Date	AQI_Bucket
count	29531	29531	24850
unique	26	2009	6
top	Ahmedabad	2020-06-26	Moderate
freq	2009	26	8829

Missing Values

[10]:
```
city.isna().sum()
```

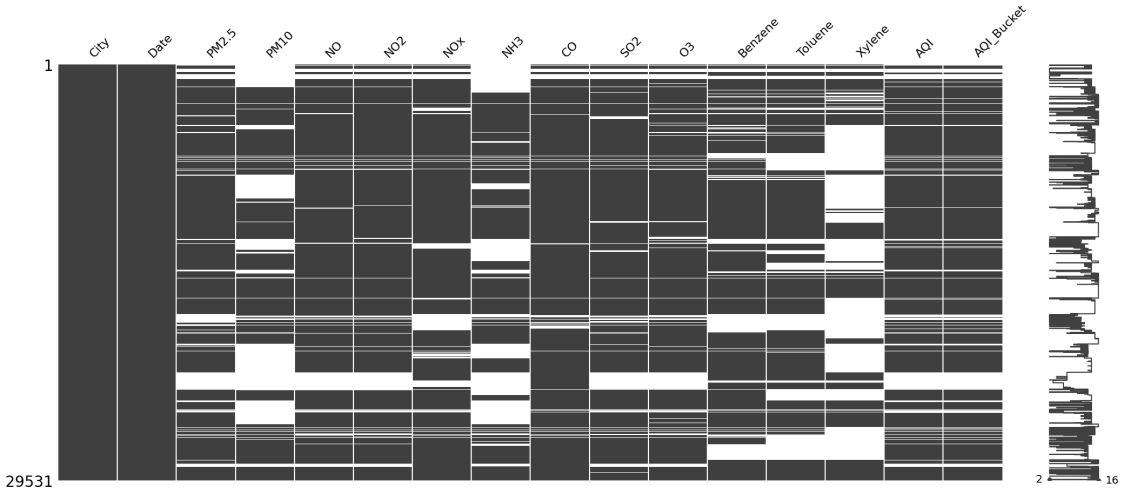
[10]:
```
City              0
Date              0
PM2.5          4598
PM10          11140
NO             3582
NO2            3585
NOx            4185
NH3           10328
CO             2059
SO2            3854
O3             4022
Benzene        5623
Toluene        8041
Xylene        18109
AQI            4681
AQI_Bucket     4681
dtype: int64
```

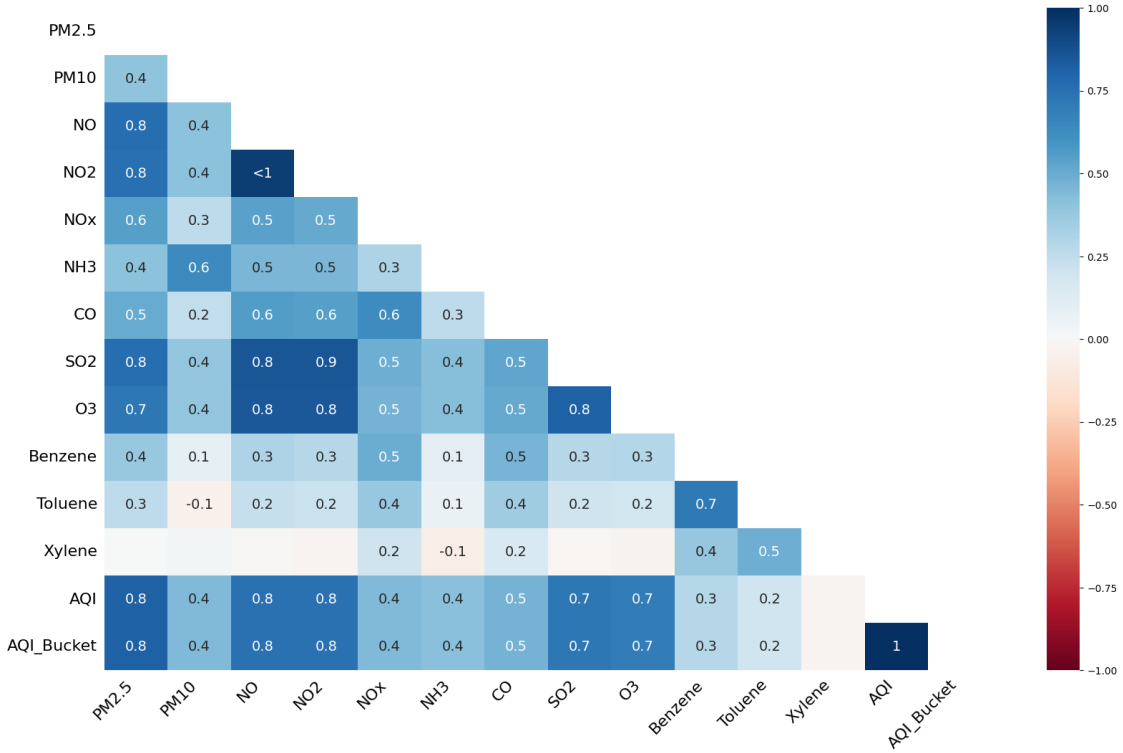
[11]:
```
((city.isna().sum())/len(city))* 100
```

[11]:
```
City           0.000000
Date           0.000000
PM2.5         15.570079
PM10          37.723071
NO            12.129626
NO2           12.139785
NOx           14.171549
NH3           34.973418
CO             6.972334
SO2           13.050692
O3            13.619586
Benzene       19.041008
Toluene       27.229014
Xylene        61.322001
AQI           15.851139
AQI_Bucket    15.851139
dtype: float64
```

[12]:
```
import missingno as msno

msno.matrix(city)
msno.heatmap(city)
```

[12]: <Axes: >





Duplicate Values

[13]:
```
city.duplicated().sum()
```

[13]:
```
np.int64(0)
```

Understanding data

[14]:
```
city.nunique()
```

[14]:
```
City             26
Date           2009
PM2.5         11716
PM10          12571
NO             5776
NO2            7404
NOx            8156
NH3            5922
CO             1779
SO2            4761
O3             7699
Benzene        1873
Toluene        3608
Xylene         1561
AQI             829
AQI_Bucket        6
dtype: int64
```

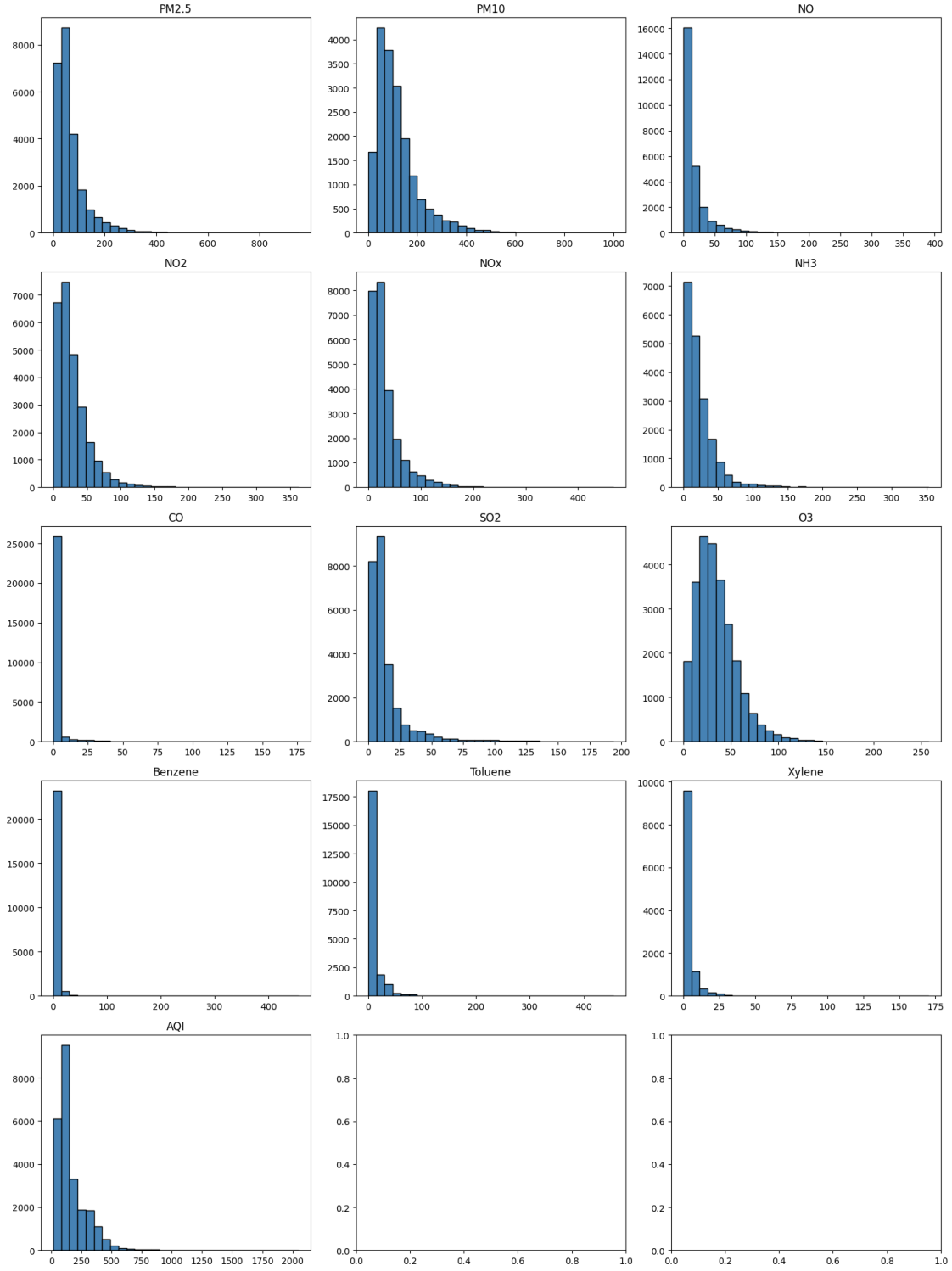
[15]:
```
city['AQI_Bucket'].unique()
```

[15]:
```
array([nan, 'Poor', 'Very Poor', 'Severe', 'Moderate', 'Satisfactory',
       'Good'], dtype=object)
```

[16]:
```
city.select_dtypes(include='number').nunique().sort_values()
```

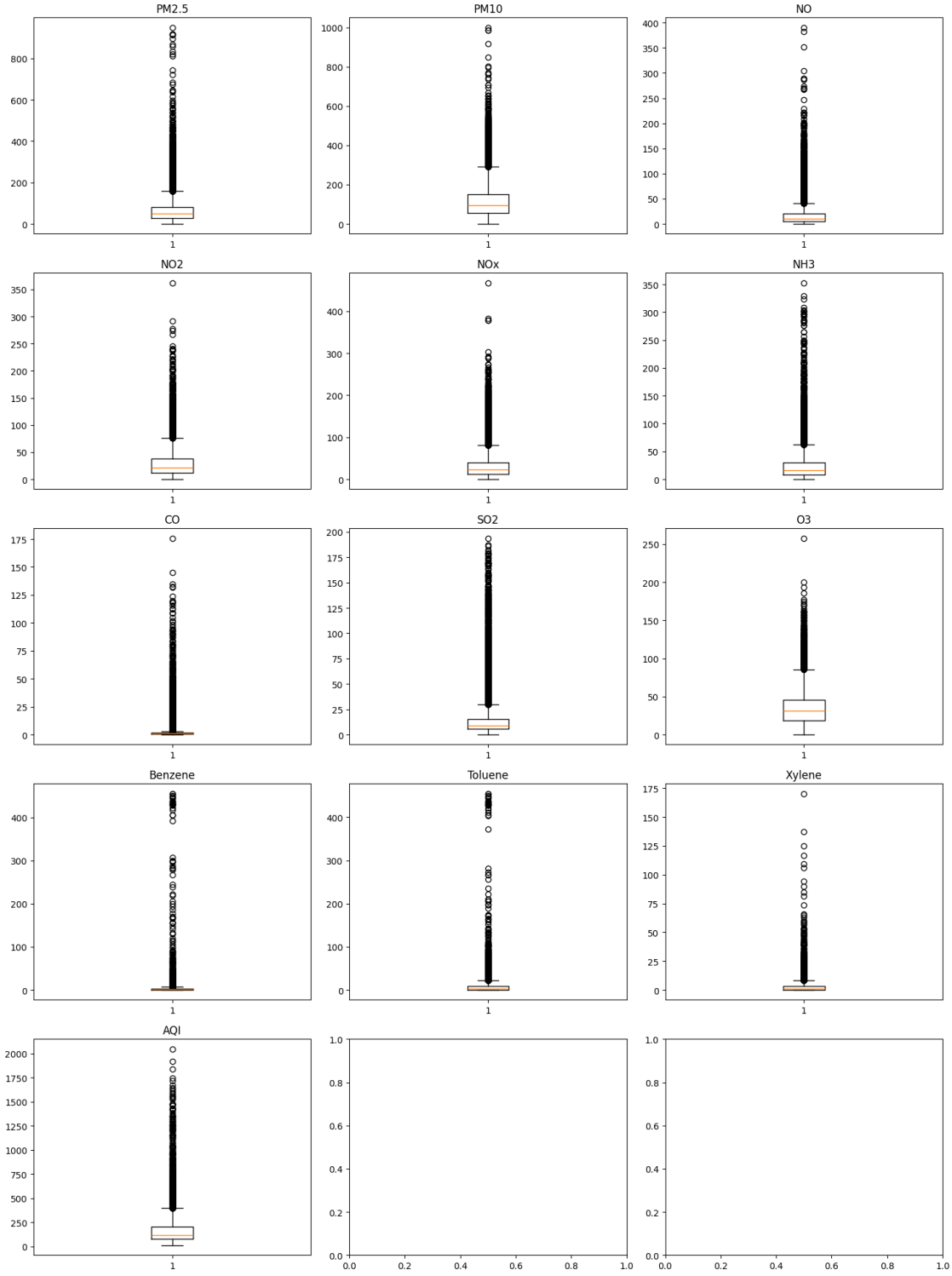
[16]:
```
AQI            829
Xylene        1561
CO            1779
Benzene       1873
Toluene       3608
SO2           4761
NO            5776
NH3           5922
NO2           7404
O3            7699
NOx           8156
PM2.5        11716
PM10         12571
dtype: int64
```

[17]:
```
import matplotlib.pyplot as plt

num_cols = city.select_dtypes(include='number').columns.tolist()
fig, axes = plt.subplots(5, 3, figsize=(15, 20))
axes = axes.flatten()

for i, col in enumerate(num_cols):
    axes[i].hist(city[col].dropna(), bins=30, color='steelblue',␣
 ↪edgecolor='black')
    axes[i].set_title(col)

plt.tight_layout()
plt.show()
```



[18]:
```
fig, axes = plt.subplots(5, 3, figsize=(15, 20))
axes = axes.flatten()
```

```
for i, col in enumerate(num_cols):
    axes[i].boxplot(city[col].dropna())
    axes[i].set_title(col)

plt.tight_layout()
plt.show()
```



Outlier Detection

[19]:
```
# IQR method
Q1 = city.select_dtypes(include='number').quantile(0.25)
Q3 = city.select_dtypes(include='number').quantile(0.75)
IQR = Q3 - Q1

outliers = ((city.select_dtypes(include='number') < (Q1 - 1.5 * IQR)) |
            (city.select_dtypes(include='number') > (Q3 + 1.5 * IQR))).sum()
print(outliers)
```

```
PM2.5       1982
PM10        1057
NO          2459
NO2         1188
NOx         1868
NH3         1015
CO          2475
SO2         2578
O3           713
Benzene     1668
Toluene     2427
Xylene      1119
AQI         1358
dtype: int64
```

Data Profile

Attribute	Detail
Shape	29,531 rows × 16 columns
Numerical columns	13 (PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene, Xylene, AQI)
Categorical columns	3 (City, Date, AQI_Bucket)
Cities covered	26
Date range	2,009 unique dates
Duplicate rows	0
Target column	AQI (with 6 categories: Good, Satisfactory, Moderate, Poor, Very Poor, Severe)

Missing Values Summary

Column	Missing Count	Missing %
Xylene	18,109	61.3%
PM10	11,140	37.7%
NH3	10,328	34.9%
Toluene	8,041	27.2%
Benzene	5,623	19.0%
AQI / AQI_Bucket	4,681	15.8%

Column	Missing Count	Missing %
PM2.5	4,598	15.5%

Outliers (IQR Method)

Column	Outlier Count
CO	2,475
SO2	2,578
NO	2,459
Toluene	2,427
PM2.5	1,982
NOx	1,868
Benzene	1,668
AQI	1,358

Distribution Shape (from Histograms) All 13 numerical columns show strong right skew — the majority of readings are low, with a long tail of extreme pollution events. AQI max reaches 2049, while the median is only 118.

Concern Xylene is missing in 61.3% of rows and may not be usable as a feature. More critically, PM2.5 and PM10 — the two most health-relevant pollutants — are missing in 15–38% of rows respectively. Every pollutant column contains significant outliers, with AQI reaching a maximum of 2,049 against a median of 118, suggesting extreme pollution events that could distort any model trained on this data without treatment.

1.1.2 Crop Dataset

Structure and Shape

[20]:
```
crop.shape
```

[20]:
```
(246091, 7)
```

[21]:
```
crop.dtypes
```

[21]:
```
State_Name        object
District_Name     object
Crop_Year          int64
Season            object
Crop              object
Area             float64
Production       float64
dtype: object
```

[22]:
```
crop.columns
```

[22]:
```
Index(['State_Name', 'District_Name', 'Crop_Year', 'Season', 'Crop', 'Area',
       'Production'],
      dtype='object')
```

Summary Statistics

[23]:
```
crop.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 246091 entries, 0 to 246090
Data columns (total 7 columns):
 #   Column         Non-Null Count   Dtype
---  ------         --------------   -----
 0   State_Name     246091 non-null  object
 1   District_Name  246091 non-null  object
 2   Crop_Year      246091 non-null  int64
 3   Season         246091 non-null  object
 4   Crop           246091 non-null  object
 5   Area           246091 non-null  float64
 6   Production     242361 non-null  float64
dtypes: float64(2), int64(1), object(4)
memory usage: 13.1+ MB
```

[24]:
```
crop.describe()
```

[24]:

	Crop_Year	Area	Production
count	246091.000000	2.460910e+05	2.423610e+05
mean	2005.643018	1.200282e+04	5.825034e+05
std	4.952164	5.052340e+04	1.706581e+07
min	1997.000000	4.000000e-02	0.000000e+00
25%	2002.000000	8.000000e+01	8.800000e+01
50%	2006.000000	5.820000e+02	7.290000e+02
75%	2010.000000	4.392000e+03	7.023000e+03
max	2015.000000	8.580100e+06	1.250800e+09

[25]:
```
crop.describe(include='object')
```

[25]:

	State_Name	District_Name	Season	Crop
count	246091	246091	246091	246091
unique	33	646	6	124
top	Uttar Pradesh	BIJAPUR	Kharif	Rice
freq	33306	945	95951	15104

Missing Values

[26]:
```
crop.isna().sum()
```

[26]:
```
State_Name          0
District_Name       0
Crop_Year           0
```

```
Season              0
Crop                0
Area                0
Production       3730
dtype: int64
```

[27]:
```
((crop.isna().sum())/len(crop))* 100
```

[27]:
```
State_Name       0.000000
District_Name    0.000000
Crop_Year        0.000000
Season           0.000000
Crop             0.000000
Area             0.000000
Production       1.515699
dtype: float64
```

Duplicate Values

[28]:
```
crop.duplicated().sum()
```

[28]:
```
np.int64(0)
```

Understanding data

[29]:
```
crop.nunique()
```

[29]:
```
State_Name          33
District_Name      646
Crop_Year           19
Season               6
Crop               124
Area             38442
Production       51627
dtype: int64
```

[30]:
```
crop.select_dtypes(include='number').nunique().sort_values()
```

[30]:
```
Crop_Year        19
Area          38442
Production    51627
dtype: int64
```

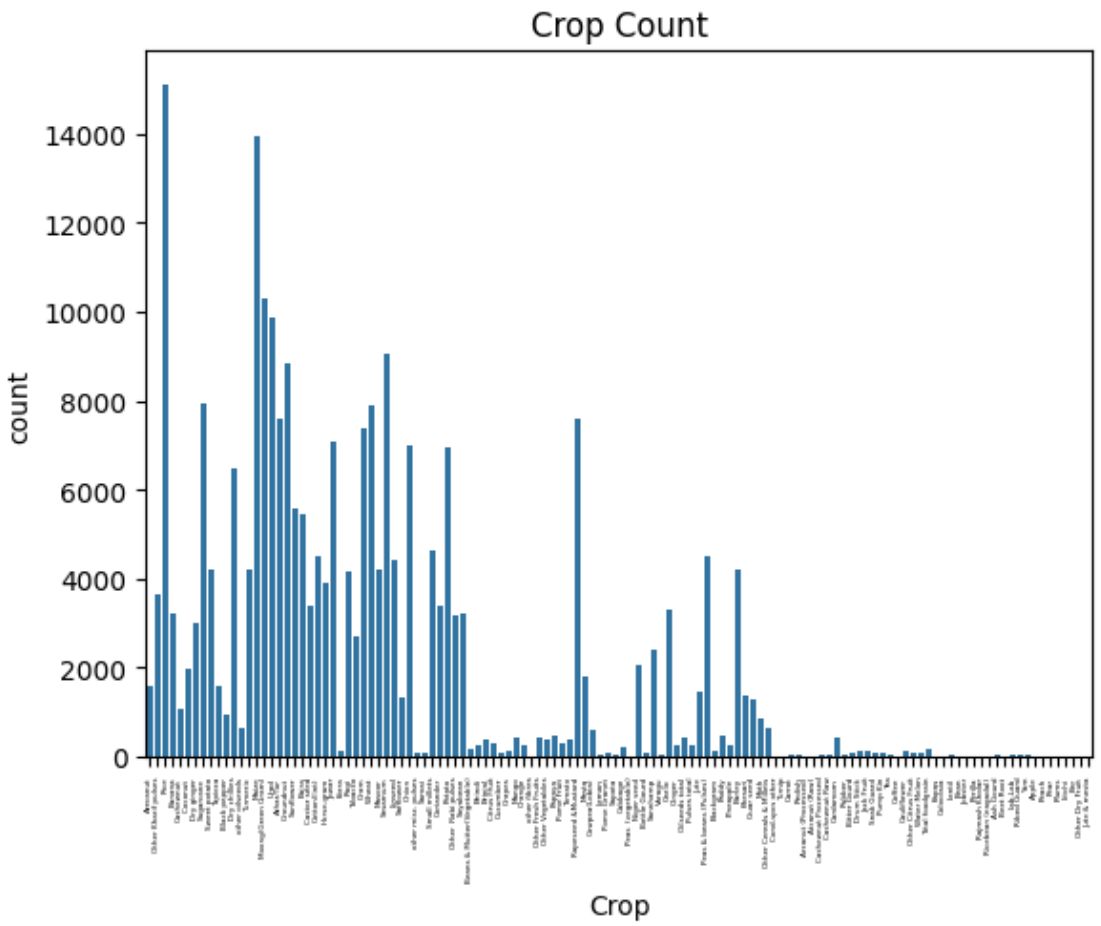
[31]:
```
crop['Season'].unique()
```

[31]:
```
array(['Kharif     ', 'Whole Year ', 'Autumn     ', 'Rabi       ',
       'Summer     ', 'Winter     '], dtype=object)
```

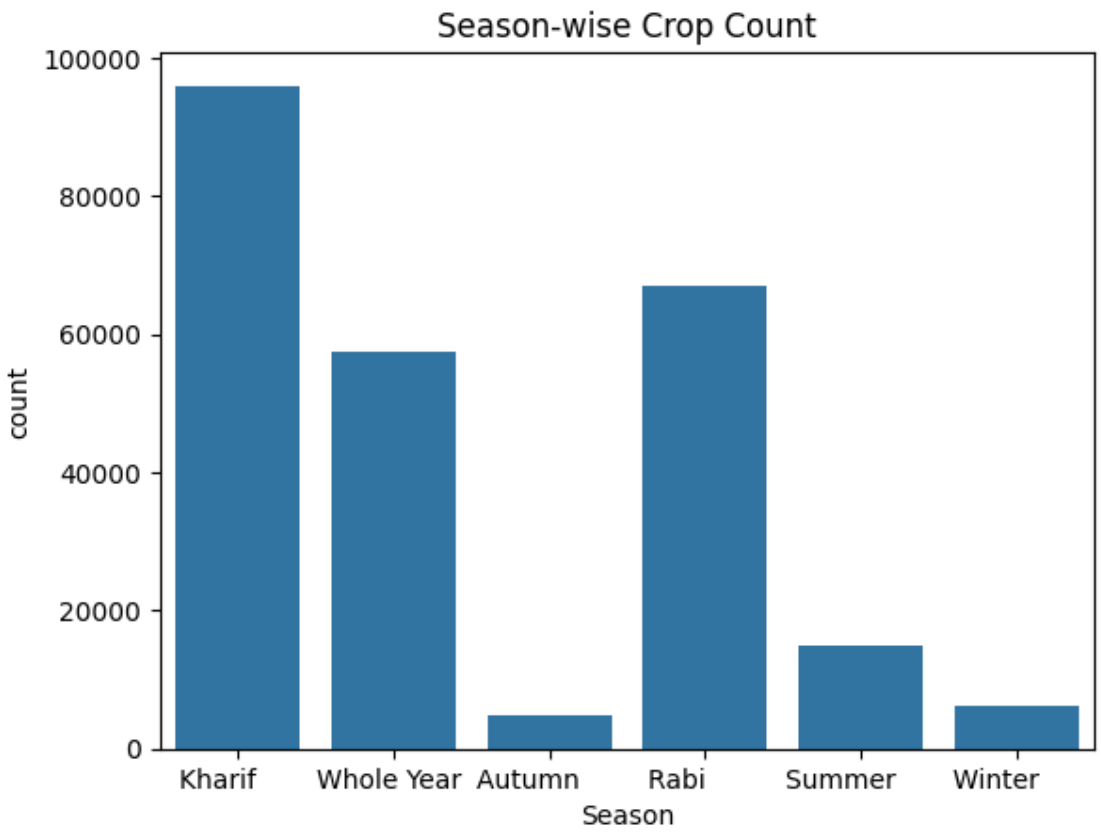
[32]:
```
crop['Crop'].unique()
```

[32]:
```
array(['Arecanut', 'Other Kharif pulses', 'Rice', 'Banana', 'Cashewnut',
       'Coconut ', 'Dry ginger', 'Sugarcane', 'Sweet potato', 'Tapioca',
       'Black pepper', 'Dry chillies', 'other oilseeds', 'Turmeric',
       'Maize', 'Moong(Green Gram)', 'Urad', 'Arhar/Tur', 'Groundnut',
       'Sunflower', 'Bajra', 'Castor seed', 'Cotton(lint)', 'Horse-gram',
       'Jowar', 'Korra', 'Ragi', 'Tobacco', 'Gram', 'Wheat', 'Masoor',
       'Sesamum', 'Linseed', 'Safflower', 'Onion', 'other misc. pulses',
       'Samai', 'Small millets', 'Coriander', 'Potato',
       'Other  Rabi pulses', 'Soyabean', 'Beans & Mutter(Vegetable)',
       'Bhindi', 'Brinjal', 'Citrus Fruit', 'Cucumber', 'Grapes', 'Mango',
       'Orange', 'other fibres', 'Other Fresh Fruits', 'Other Vegetables',
       'Papaya', 'Pome Fruit', 'Tomato', 'Rapeseed &Mustard', 'Mesta',
       'Cowpea(Lobia)', 'Lemon', 'Pome Granet', 'Sapota', 'Cabbage',
       'Peas  (vegetable)', 'Niger seed', 'Bottle Gourd', 'Sannhamp',
       'Varagu', 'Garlic', 'Ginger', 'Oilseeds total', 'Pulses total',
       'Jute', 'Peas & beans (Pulses)', 'Blackgram', 'Paddy', 'Pineapple',
       'Barley', 'Khesari', 'Guar seed', 'Moth',
       'Other Cereals & Millets', 'Cond-spcs other', 'Turnip', 'Carrot',
       'Redish', 'Arcanut (Processed)', 'Atcanut (Raw)',
       'Cashewnut Processed', 'Cashewnut Raw', 'Cardamom', 'Rubber',
       'Bitter Gourd', 'Drum Stick', 'Jack Fruit', 'Snak Guard',
       'Pump Kin', 'Tea', 'Coffee', 'Cauliflower', 'Other Citrus Fruit',
       'Water Melon', 'Total foodgrain', 'Kapas', 'Colocosia', 'Lentil',
       'Bean', 'Jobster', 'Perilla', 'Rajmash Kholar',
       'Ricebean (nagadal)', 'Ash Gourd', 'Beet Root', 'Lab-Lab',
       'Ribed Guard', 'Yam', 'Apple', 'Peach', 'Pear', 'Plums', 'Litchi',
       'Ber', 'Other Dry Fruit', 'Jute & mesta'], dtype=object)
```

[33]:
```
import seaborn as sns
sns.countplot(x='Crop', data = crop)
plt.title("Crop Count")
plt.xticks(rotation=90, fontsize=3)
plt.show()
```

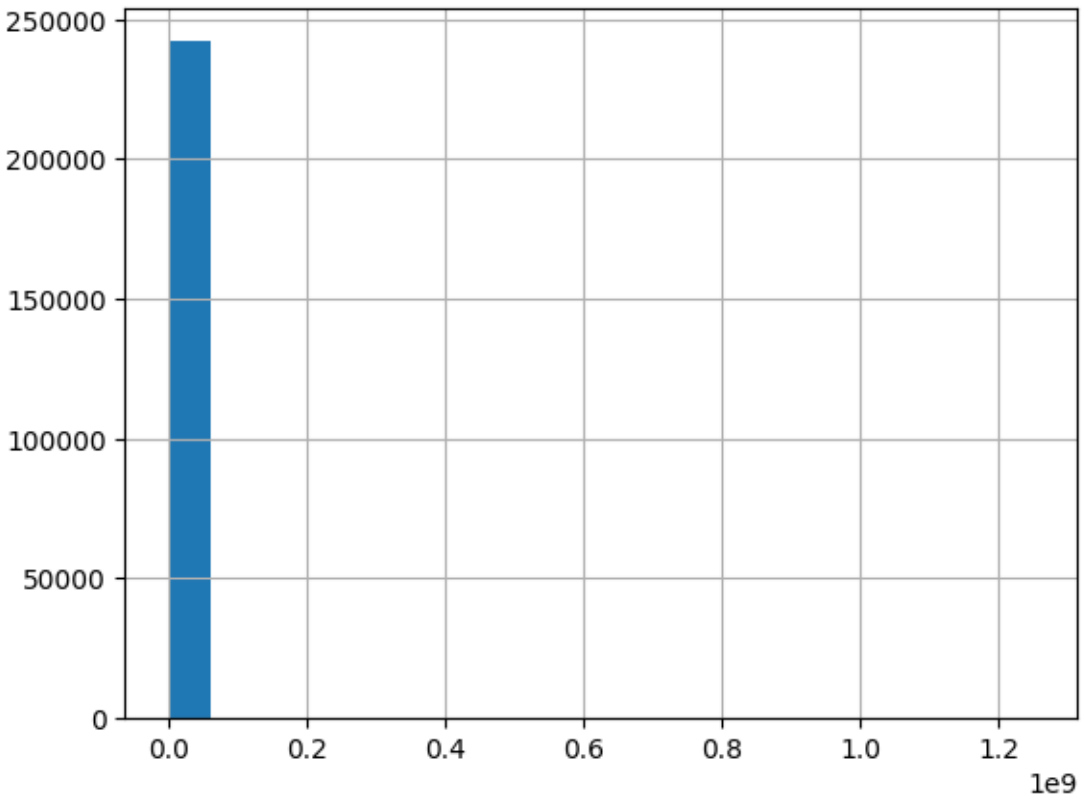


[34]:
```
import seaborn as sns
sns.countplot(x='Season', data = crop)
plt.title("Season-wise Crop Count")
plt.show()
```



[35]:
```
crop['Production'].hist(bins=20)
```

[35]: <Axes: >



Issues with Production distribution must be handled later through normalisation.

Outlier Detection

[36]:
```
# IQR method
Q1 = crop.select_dtypes(include='number').quantile(0.25)
Q3 = crop.select_dtypes(include='number').quantile(0.75)
IQR = Q3 - Q1

outliers = ((crop.select_dtypes(include='number') < (Q1 - 1.5 * IQR)) |
            (crop.select_dtypes(include='number') > (Q3 + 1.5 * IQR))).sum()
print(outliers)
```

```
Crop_Year         0
Area          40705
Production    42390
dtype: int64
```

Data Profile

Attribute	Detail
Shape	246,091 rows × 7 columns

Attribute	Detail
Numerical columns	3 (Crop_Year, Area, Production)
Categorical columns	4 (State_Name, District_Name, Season, Crop)
States covered	33
Districts covered	646
Year range	1997 – 2015 (19 unique years)
Unique crops	124
Unique seasons	6
Duplicate rows	0

Missing Values Summary

Column	Missing Count	Missing %
Production	3,730	1.5%
All others	0	0%

1.1.3 Outliers (IQR Method)

Column	Outlier Count
Area	40,705
Production	42,390

Key Observations from Distributions

- **Season:** Kharif dominates (~96,000 records), followed by Rabi (~67,000). Autumn and Winter are sparsely represented.
- **Crop:** Rice is the most recorded crop (15,104 records). Many crops appear very rarely — potential class imbalance issue.
- **Area & Production:** Extreme right skew with massive outlier counts. Mean production (582,503) is astronomically higher than median (729), indicating a few large-scale crops pulling the average up heavily.

Concern Area and Production have over 40,000 outliers each — nearly 16% of all rows. This is not noise; it reflects genuine variation between small subsistence farms and large commercial operations. However, the gap between mean (582,503) and median (729) in Production is extreme and will severely distort any ML model if left untreated. A log transformation will likely be necessary before modelling. Additionally, Season values contain trailing spaces (e.g. 'Kharif') which must be cleaned before the dataset can be merged with the city file.

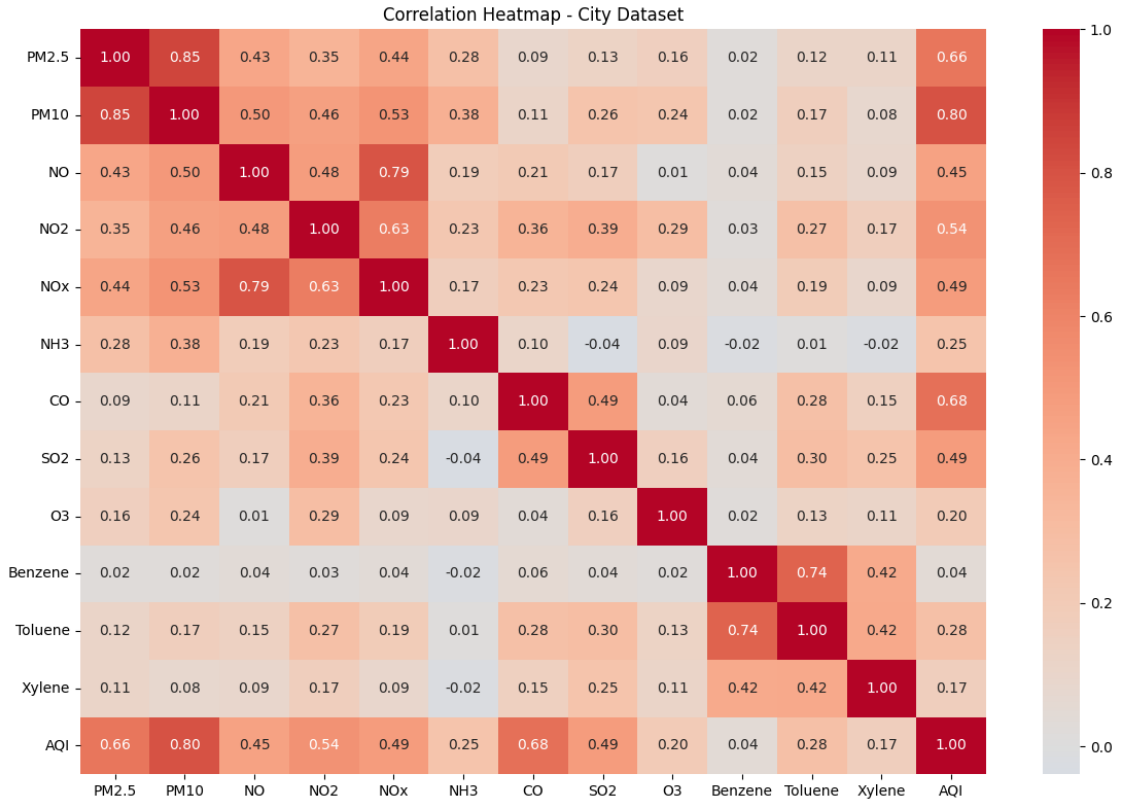
1.2 Task 2 : Handling Missing Values

[37]:
```
print(city.isna().sum()[city.isna().sum() > 0]/len(city)*100, "\n")
print(crop.isna().sum()[crop.isna().sum() > 0]/len(crop)*100)
```

```
PM2.5          15.570079
PM10           37.723071
NO             12.129626
NO2            12.139785
NOx            14.171549
NH3            34.973418
CO              6.972334
SO2            13.050692
O3             13.619586
Benzene        19.041008
Toluene        27.229014
Xylene         61.322001
AQI            15.851139
AQI_Bucket     15.851139
dtype: float64

Production     1.515699
dtype: float64
```

Dropping Columns

[38]:
```
corr = city.select_dtypes(include='number').corr()

plt.figure(figsize=(12, 8))
sns.heatmap(corr, annot=True, fmt='.2f', cmap='coolwarm', center=0)
plt.title('Correlation Heatmap - City Dataset')
plt.tight_layout()
plt.show()
```



Before dropping columns with high missing values, a correlation heatmap was examined to assess each column's relationship with AQI. Xylene (0.17) and Benzene (0.04) showed negligible correlation with AQI and were dropped. PM10 (0.80), CO (0.68), and PM2.5 (0.66) showed the strongest predictive signal and were retained despite moderate missing percentages.

[39]:
```
cols_to_drop = ['Benzene', 'Xylene', 'Toluene']
city.drop(columns=cols_to_drop, inplace=True)
print("Columns remaining:", city.columns.tolist())
print("Shape after drop:", city.shape)
```

```
Columns remaining: ['City', 'Date', 'PM2.5', 'PM10', 'NO', 'NO2', 'NOx', 'NH3',
'CO', 'SO2', 'O3', 'AQI', 'AQI_Bucket']
Shape after drop: (29531, 13)
```

Imputing Columns

[40]:
```
print("Before imputation:")
print(city.isna().sum()[city.isna().sum() > 0]/len(city)*100)
```

```
Before imputation:
PM2.5          15.570079
PM10           37.723071
NO             12.129626
NO2            12.139785
```

```
NOx            14.171549
NH3            34.973418
CO              6.972334
SO2            13.050692
O3             13.619586
AQI            15.851139
AQI_Bucket     15.851139
dtype: float64
```

Imputing with Median & Mode Before imputing, the distribution of each column was examined using histograms and `.describe()` in Task 1.

All pollutant columns (PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, AQI) showed strong **right-skewed distributions** — the bulk of readings are low, but a few extreme pollution events push the tail far to the right.

In a skewed distribution: - The **mean** gets pulled toward the outliers and no longer represents a typical value - The **median** is resistant to outliers and better represents what a "normal" day looks like

For example, AQI has a mean of 166 but a median of 118. If we filled missing values with 166, we would be overstating pollution on those days. Filling with 118 is more honest.

Rule applied: - Skewed numeric columns → impute with **median** - Normal distribution → impute with **mean** (not applicable here) - Categorical column (AQI_Bucket) → impute with **mode** (most frequent category)

Columns Dropped Before Imputation Benzene, Toluene, and Xylene were dropped based on two reasons: 1. Low correlation with AQI (0.04, 0.28, 0.17 respectively) 2. Xylene additionally had 61.3% missing values

Crop Dataset Production had only 1.5% missing values. The distribution is heavily right-skewed (mean = 582,503 vs median = 729), so median imputation was applied.

[41]:
```
cols_to_impute = ['PM2.5', 'PM10', 'NO', 'NO2', 'NOx', 'NH3', 'CO', 'SO2',␣
↪'O3', 'AQI']
```

[42]:
```
for col in cols_to_impute:
    city[col].fillna(city[col].median(), inplace=True)
```

```
C:\Users\neha_\AppData\Local\Temp\ipykernel_39876\1725500912.py:2:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series
through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work
because the intermediate object on which we are setting values always behaves as
a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value)
instead, to perform the operation inplace on the original object.
```

```
  city[col].fillna(city[col].median(), inplace=True)
C:\Users\neha_\AppData\Local\Temp\ipykernel_39876\1725500912.py:2:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series
through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work
because the intermediate object on which we are setting values always behaves as
a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value)
instead, to perform the operation inplace on the original object.


  city[col].fillna(city[col].median(), inplace=True)
C:\Users\neha_\AppData\Local\Temp\ipykernel_39876\1725500912.py:2:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series
through chained assignment using an inplace method.
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For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value)
instead, to perform the operation inplace on the original object.


  city[col].fillna(city[col].median(), inplace=True)
C:\Users\neha_\AppData\Local\Temp\ipykernel_39876\1725500912.py:2:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series
through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work
because the intermediate object on which we are setting values always behaves as
a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value)
instead, to perform the operation inplace on the original object.


  city[col].fillna(city[col].median(), inplace=True)
C:\Users\neha_\AppData\Local\Temp\ipykernel_39876\1725500912.py:2:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series
through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work
because the intermediate object on which we are setting values always behaves as
a copy.
```
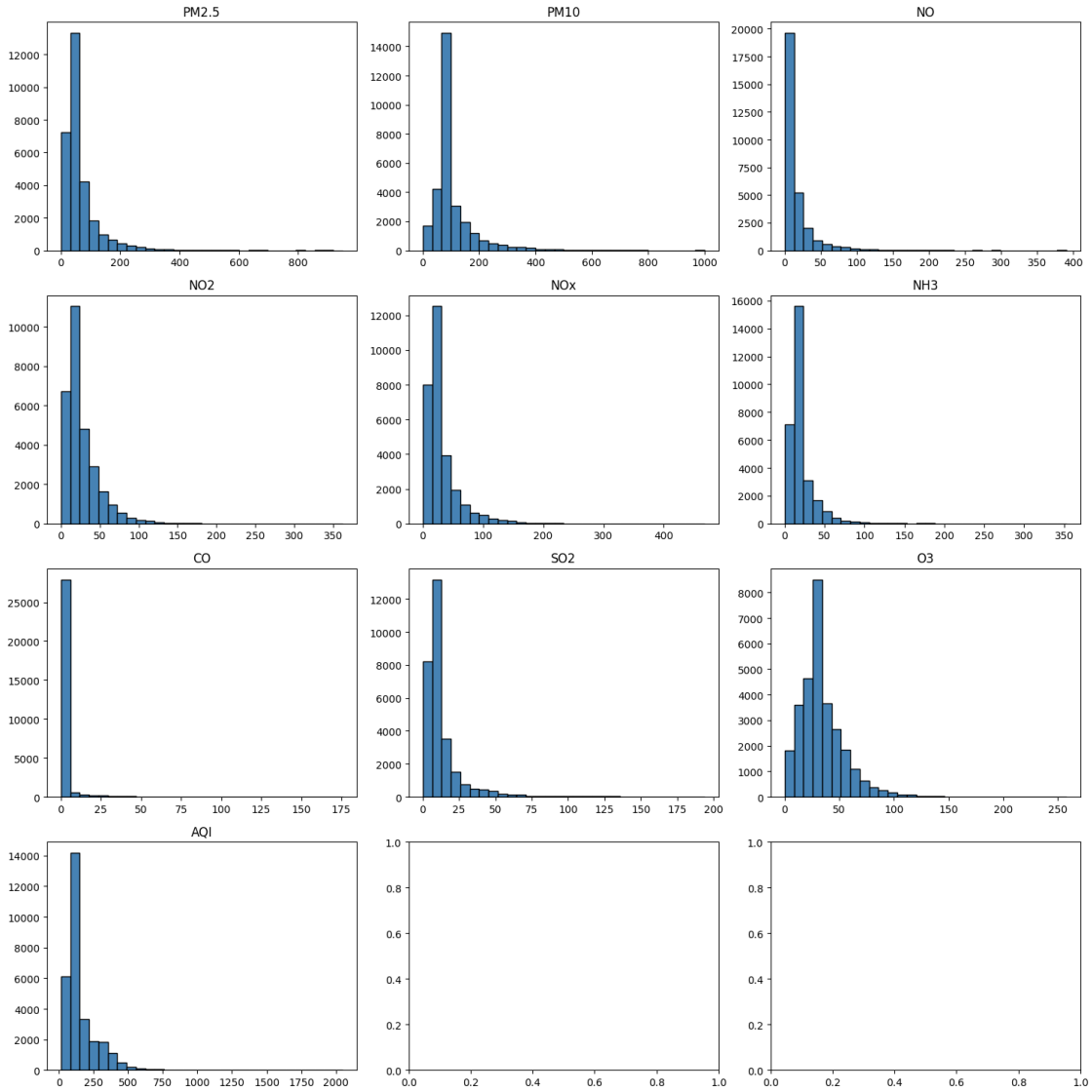

```
For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value)
instead, to perform the operation inplace on the original object.


  city[col].fillna(city[col].median(), inplace=True)
C:\Users\neha_\AppData\Local\Temp\ipykernel_39876\1725500912.py:2:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series
through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work
because the intermediate object on which we are setting values always behaves as
a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value)
instead, to perform the operation inplace on the original object.


  city[col].fillna(city[col].median(), inplace=True)
C:\Users\neha_\AppData\Local\Temp\ipykernel_39876\1725500912.py:2:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series
through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work
because the intermediate object on which we are setting values always behaves as
a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value)
instead, to perform the operation inplace on the original object.


  city[col].fillna(city[col].median(), inplace=True)
C:\Users\neha_\AppData\Local\Temp\ipykernel_39876\1725500912.py:2:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series
through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work
because the intermediate object on which we are setting values always behaves as
a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value)
instead, to perform the operation inplace on the original object.


  city[col].fillna(city[col].median(), inplace=True)
C:\Users\neha_\AppData\Local\Temp\ipykernel_39876\1725500912.py:2:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series
through chained assignment using an inplace method.
```

```
The behavior will change in pandas 3.0. This inplace method will never work
because the intermediate object on which we are setting values always behaves as
a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value)
instead, to perform the operation inplace on the original object.


  city[col].fillna(city[col].median(), inplace=True)
C:\Users\neha_\AppData\Local\Temp\ipykernel_39876\1725500912.py:2:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series
through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work
because the intermediate object on which we are setting values always behaves as
a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value)
instead, to perform the operation inplace on the original object.


  city[col].fillna(city[col].median(), inplace=True)
```

[43]:
```
cols_to_impute = city.select_dtypes(include='number').columns.tolist()
fig, axes = plt.subplots(4, 3, figsize=(15, 15))
axes = axes.flatten()

for i, col in enumerate(cols_to_impute):
    axes[i].hist(city[col].dropna(), bins=30, color='steelblue',␣
 ↪edgecolor='black')
    axes[i].set_title(col)

plt.tight_layout()
plt.show()
```



Skewness remains after imputation.Imputation handled correctly.

[44]:
```
print(city.isna().sum()[city.isna().sum() > 0]/len(city)*100)
```

```
AQI_Bucket    15.851139
dtype: float64
```

[45]:
```
city['AQI_Bucket'].fillna(city['AQI_Bucket'].mode()[0], inplace=True)
```

```
C:\Users\neha_\AppData\Local\Temp\ipykernel_39876\2134353856.py:1:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series
through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work
because the intermediate object on which we are setting values always behaves as
```

```
a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value)
instead, to perform the operation inplace on the original object.


  city['AQI_Bucket'].fillna(city['AQI_Bucket'].mode()[0], inplace=True)
```

[46]:
```
print(city.isna().sum()[city.isna().sum() > 0]/len(city)*100)
```

```
Series([], dtype: float64)
```

City Dataset has no missing values.

[47]:
```
print(crop.isna().sum()[crop.isna().sum() > 0]/len(crop)*100)
```

```
Production    1.515699
dtype: float64
```

[48]:
```
crop.head()
```

[48]:

	State_Name	District_Name	Crop_Year	Season	Crop	Area	Production
0	Andaman and Nicobar Islands	NICOBARS	2000	Kharif	Arecanut	1254.0	2000.0
1	Andaman and Nicobar Islands	NICOBARS	2000	Kharif	Other Kharif pulses	2.0	1.0
2	Andaman and Nicobar Islands	NICOBARS	2000	Kharif	Rice	102.0	321.0
3	Andaman and Nicobar Islands	NICOBARS	2000	Whole Year	Banana	176.0	641.0
4	Andaman and Nicobar Islands	NICOBARS	2000	Whole Year	Cashewnut	720.0	165.0

[49]:
```
pd.set_option('display.float_format', '{:.2f}'.format)
crop['Production'].describe()
```

[49]:
```
count       242361.00
mean        582503.44
std       17065813.17
min              0.00
25%             88.00
50%            729.00
75%           7023.00
max     1250800000.00
Name: Production, dtype: float64
```

Mean = 582503; Median = 729

Mean> Median = Skewed

Therefore Production column imputed with median

[50]:
```
crop['Production'].fillna(crop['Production'].median(), inplace=True)
```

```
C:\Users\neha_\AppData\Local\Temp\ipykernel_39876\2561813069.py:1:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series
through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work
because the intermediate object on which we are setting values always behaves as
a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value)
instead, to perform the operation inplace on the original object.


  crop['Production'].fillna(crop['Production'].median(), inplace=True)
```

[51]:
```
print(crop.isna().sum()[crop.isna().sum() > 0]/len(crop)*100)
```

```
Series([], dtype: float64)
```

Crop Dataset has no missing values.

[52]:
```
print("City nulls remaining:", city.isna().sum().sum())
print("Crop nulls remaining:", crop.isna().sum().sum())
```

```
City nulls remaining: 0
Crop nulls remaining: 0
```

Columns with missing data were treated based on two factors: the percentage missing and the distribution shape. Xylene was dropped as 61.3% of values were absent, making it unreliable for modelling. All remaining numeric pollutant columns were imputed using the column median rather than mean, because their distributions are strongly right-skewed — the mean is inflated by extreme pollution events and would not represent a typical missing reading accurately. AQI_Bucket was imputed with the mode as it is categorical. In the crop dataset, Production had only 1.5% missing values and was imputed with the median.

1.3 Task 3 : Improving Data Quality

Trailing Spaces During profiling, Season values showed trailing whitespace (e.g. 'Kharif'). This causes silent mismatches — 'Kharif' and 'Kharif' are treated as different categories by Python, which would break groupby operations and any future merge. Stripping whitespace from all categorical columns ensures consistent matching.

[53]:
```
for col in crop.select_dtypes(include='object').columns:
    crop[col] = crop[col].str.strip()

for col in city.select_dtypes(include='object').columns:
```

```
    city[col] = city[col].str.strip()

print(crop['Season'].unique())
print(city['AQI_Bucket'].unique())
```

```
['Kharif' 'Whole Year' 'Autumn' 'Rabi' 'Summer' 'Winter']
['Moderate' 'Poor' 'Very Poor' 'Severe' 'Satisfactory' 'Good']
```

Capitalisation Inconsistent casing (e.g. 'uttar pradesh' vs 'Uttar Pradesh') would cause a merge to miss matching rows silently — no error is thrown, records are just lost. Title case is applied uniformly so state names match exactly across both files.

[54]:
```
crop['State_Name'] = crop['State_Name'].str.title()
city['City'] = city['City'].str.title()

# Verify
print(sorted(crop['State_Name'].unique()))
print(sorted(city['City'].unique()))
```

```
['Andaman And Nicobar Islands', 'Andhra Pradesh', 'Arunachal Pradesh', 'Assam',
'Bihar', 'Chandigarh', 'Chhattisgarh', 'Dadra And Nagar Haveli', 'Goa',
'Gujarat', 'Haryana', 'Himachal Pradesh', 'Jammu And Kashmir', 'Jharkhand',
'Karnataka', 'Kerala', 'Madhya Pradesh', 'Maharashtra', 'Manipur', 'Meghalaya',
'Mizoram', 'Nagaland', 'Odisha', 'Puducherry', 'Punjab', 'Rajasthan', 'Sikkim',
'Tamil Nadu', 'Telangana', 'Tripura', 'Uttar Pradesh', 'Uttarakhand', 'West
Bengal']
['Ahmedabad', 'Aizawl', 'Amaravati', 'Amritsar', 'Bengaluru', 'Bhopal',
'Brajrajnagar', 'Chandigarh', 'Chennai', 'Coimbatore', 'Delhi', 'Ernakulam',
'Gurugram', 'Guwahati', 'Hyderabad', 'Jaipur', 'Jorapokhar', 'Kochi', 'Kolkata',
'Lucknow', 'Mumbai', 'Patna', 'Shillong', 'Talcher', 'Thiruvananthapuram',
'Visakhapatnam']
```

Date format The Date column is stored as object (string) dtype. Keeping it as a string prevents any time-based operations — grouping by year, extracting month, plotting trends over time. Converting to datetime unlocks all of these for later tasks.

[55]:
```
# Check what it currently is
print(city['Date'].dtype)

# Convert to datetime
city['Date'] = pd.to_datetime(city['Date'])

# Verify
print(city['Date'].dtype)
print(city['Date'].head())
```

```
object
datetime64[ns]
0   2015-01-01
```

```
1   2015-01-02
2   2015-01-03
3   2015-01-04
4   2015-01-05
Name: Date, dtype: datetime64[ns]
```

[56]:
```
print("City date range:")
print("From:", city['Date'].min())
print("To:", city['Date'].max())

print("\nCrop year range:")
print("From:", crop['Crop_Year'].min())
print("To:", crop['Crop_Year'].max())
```

```
City date range:
From: 2015-01-01 00:00:00
To: 2020-07-01 00:00:00

Crop year range:
From: 1997
To: 2015
```

The city dataset covers 2015–2023 while the crop dataset covers 1997–2015. The two datasets share only one overlapping year (2015), which significantly limits the scope of any joined analysis. This is a structural limitation of the data and cannot be resolved through cleaning — it must be acknowledged when interpreting results in later tasks.

State mismatches Since both the datasets need to be joined based on state, evaluating respective location columns

[57]:
```
print(sorted(city['City'].unique()))
```

```
['Ahmedabad', 'Aizawl', 'Amaravati', 'Amritsar', 'Bengaluru', 'Bhopal',
'Brajrajnagar', 'Chandigarh', 'Chennai', 'Coimbatore', 'Delhi', 'Ernakulam',
'Gurugram', 'Guwahati', 'Hyderabad', 'Jaipur', 'Jorapokhar', 'Kochi', 'Kolkata',
'Lucknow', 'Mumbai', 'Patna', 'Shillong', 'Talcher', 'Thiruvananthapuram',
'Visakhapatnam']
```

[58]:
```
print(sorted(crop['State_Name'].unique()))
```

```
['Andaman And Nicobar Islands', 'Andhra Pradesh', 'Arunachal Pradesh', 'Assam',
'Bihar', 'Chandigarh', 'Chhattisgarh', 'Dadra And Nagar Haveli', 'Goa',
'Gujarat', 'Haryana', 'Himachal Pradesh', 'Jammu And Kashmir', 'Jharkhand',
'Karnataka', 'Kerala', 'Madhya Pradesh', 'Maharashtra', 'Manipur', 'Meghalaya',
'Mizoram', 'Nagaland', 'Odisha', 'Puducherry', 'Punjab', 'Rajasthan', 'Sikkim',
'Tamil Nadu', 'Telangana', 'Tripura', 'Uttar Pradesh', 'Uttarakhand', 'West
Bengal']
```

[59]:
```
crop.head()
```

[59]:
```
                    State_Name District_Name  Crop_Year      Season  \
0  Andaman And Nicobar Islands      NICOBARS       2000      Kharif
1  Andaman And Nicobar Islands      NICOBARS       2000      Kharif
2  Andaman And Nicobar Islands      NICOBARS       2000      Kharif
3  Andaman And Nicobar Islands      NICOBARS       2000  Whole Year
4  Andaman And Nicobar Islands      NICOBARS       2000  Whole Year

                  Crop    Area  Production
0             Arecanut 1254.00     2000.00
1  Other Kharif pulses    2.00        1.00
2                 Rice  102.00      321.00
3               Banana  176.00      641.00
4            Cashewnut  720.00      165.00
```

No State columns in City Dataset

[60]:
```
print("City rows:", len(city))
print("Crop rows:", len(crop))
```

```
City rows: 29531
Crop rows: 246091
```

Mapping Cities to State

[61]:
```
city_to_state = {
    'Ahmedabad': 'Gujarat',
    'Aizawl': 'Mizoram',
    'Amaravati': 'Andhra Pradesh',
    'Amritsar': 'Punjab',
    'Bengaluru': 'Karnataka',
    'Bhopal': 'Madhya Pradesh',
    'Brajrajnagar': 'Odisha',
    'Chandigarh': 'Chandigarh',
    'Chennai': 'Tamil Nadu',
    'Coimbatore': 'Tamil Nadu',
    'Delhi': 'Delhi',
    'Ernakulam': 'Kerala',
    'Gurugram': 'Haryana',
    'Guwahati': 'Assam',
    'Hyderabad': 'Telangana',
    'Jaipur': 'Rajasthan',
    'Jorapokhar': 'Jharkhand',
    'Kochi': 'Kerala',
    'Kolkata': 'West Bengal',
    'Lucknow': 'Uttar Pradesh',
    'Mumbai': 'Maharashtra',
    'Patna': 'Bihar',
    'Shillong': 'Meghalaya',
    'Talcher': 'Odisha',
```

```
    'Thiruvananthapuram': 'Kerala',
    'Visakhapatnam': 'Andhra Pradesh'
}

city['State'] = city['City'].map(city_to_state)
print("Unmapped cities:", city[city['State'].isna()]['City'].unique())
```

```
Unmapped cities: []
```

[62]:
```
state_fixes = {
    'Andaman And Nicobar Islands': 'Andaman and Nicobar Islands',
    'Dadra And Nagar Haveli': 'Dadra and Nagar Haveli',
    'Jammu And Kashmir': 'Jammu and Kashmir',
}

crop['State_Name'] = crop['State_Name'].replace(state_fixes)

crop.rename(columns={'State_Name': 'State'}, inplace=True)
```

[63]:
```
crop_states = set(crop['State'].unique())
city_states = set(city['State'].dropna().unique())

print("In city but not in crop:", city_states - crop_states)
print("In crop but not in city:", crop_states - city_states)
```

```
In city but not in crop: {'Delhi'}
In crop but not in city: {'Dadra and Nagar Haveli', 'Jammu and Kashmir',
'Himachal Pradesh', 'Arunachal Pradesh', 'Nagaland', 'Tripura', 'Puducherry',
'Andaman and Nicobar Islands', 'Manipur', 'Uttarakhand', 'Goa', 'Sikkim',
'Chhattisgarh'}
```

[64]:
```
print("Before dedup:")
print("City duplicates:", city.duplicated().sum())
print("Crop duplicates:", crop.duplicated().sum())
```

```
Before dedup:
City duplicates: 0
Crop duplicates: 0
```

The city dataset contains city-level data with no State column, while the crop dataset contains state-level data. To merge them later, a common key is needed. Additionally, state names across both files may have inconsistencies that would cause rows to be silently dropped during a merge.

Inconsistencies Found and Fixed

Inconsistency	Found In	Fix Applied
City file has no State column	city	Mapped each city to its state manually

Inconsistency	Found In	Fix Applied
'Andaman And Nicobar Islands' (capital And)	crop	Replaced with 'Andaman and Nicobar Islands'
'Dadra And Nagar Haveli' (capital And)	crop	Replaced with 'Dadra and Nagar Haveli'
'Jammu And Kashmir' (capital And)	crop	Replaced with 'Jammu and Kashmir'
Trailing spaces in Season, State_Name	crop	Stripped using .str.strip() in Task 2

Duplicates Both files showed 0 duplicate rows after the cleaning steps above. Row counts remained unchanged, confirming no data was lost during standardisation.

1.4 Task 4 : Where do most cities actually sit on the AQI scale?

The Air Quality Index (AQI) translates complex air pollution data into a single, easy-to-understand number on a scale of 0 to 500. It measures key pollutants—primarily Particulate Matter ((PM_{2.5})and PM), Ozone, Nitrogen Dioxide, Sulfur Dioxide, and Carbon Monoxide—indicating the associated health risks.

AQI Index Interpretation: - 0–50 (Good): Air is safe. Enjoy outdoor activities. - 51–100 (Moderate): Safe for most. Sensitive people should monitor symptoms. - 101–150 (Unhealthy for Sensitive Groups): Risks for children, elderly, and those with lung/heart conditions. - 151–200 (Unhealthy): Everyone may feel effects. Cut back on heavy outdoor exercise.201–300 (Very Unhealthy): Health alert. Avoid spending time outdoors. - 301+ (Hazardous): Health emergency. Everyone must stay indoors.

Questions to answer: - Are most Indian cities moderately polluted, or is the problem concentrated in just a few places? - Whether the average AQI is a fair number to report publicly, or whether extreme cities are pulling it up unfairly.

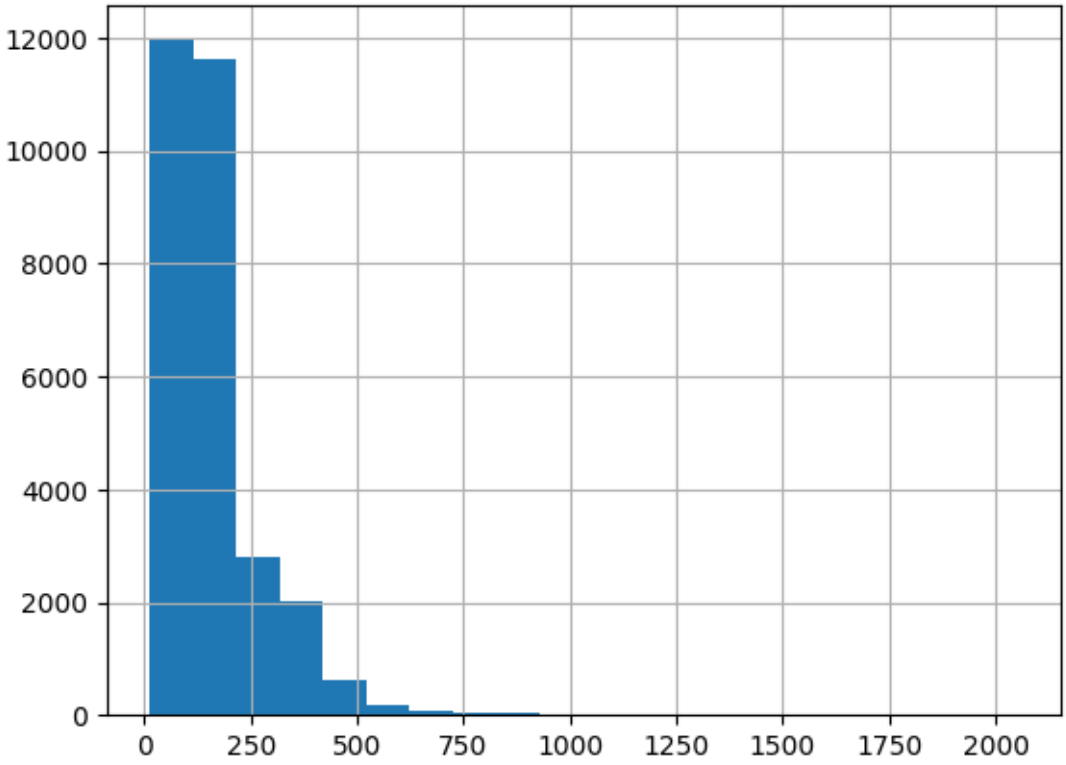
[65]:
```
city['AQI'].describe()
```

[65]:
```
count   29531.00
mean      158.78
std       130.27
min        13.00
25%        88.00
50%       118.00
75%       179.00
max      2049.00
Name: AQI, dtype: float64
```

Mean(158.78)> Median (118). Therefore the dataset is right-skewed, which means that extreme pollution events are pulling the average up.

[66]:
```
city['AQI'].hist(bins=20)
```

[66]: <Axes: >



The data is skewed as shown in the visualisation and the range of data is from 13 to 2049. AQI index of 2041 is abnormally high, therefore outliers exist.

[67]:
```
Q1 = np.quantile(city['AQI'],0.25)
Q3= np.quantile(city['AQI'],0.75)

IQR= Q3-Q1

calc_min = Q1-1.5*IQR
calc_max = Q3+1.5*IQR

outliers = city[(city['AQI'] < calc_min) | (city['AQI'] > calc_max)]
```

[68]:
```
print("Total outlier rows:", len(outliers))
print("Percentage of data:", round(len(outliers)/len(city)*100, 2), "%")
print("\nAQI range of outliers:")
print("Min:", outliers['AQI'].min())
print("Max:", outliers['AQI'].max())
print("\nIQR boundaries:")
print("Lower bound:", calc_min)
```

```
print("Upper bound:", calc_max)
```

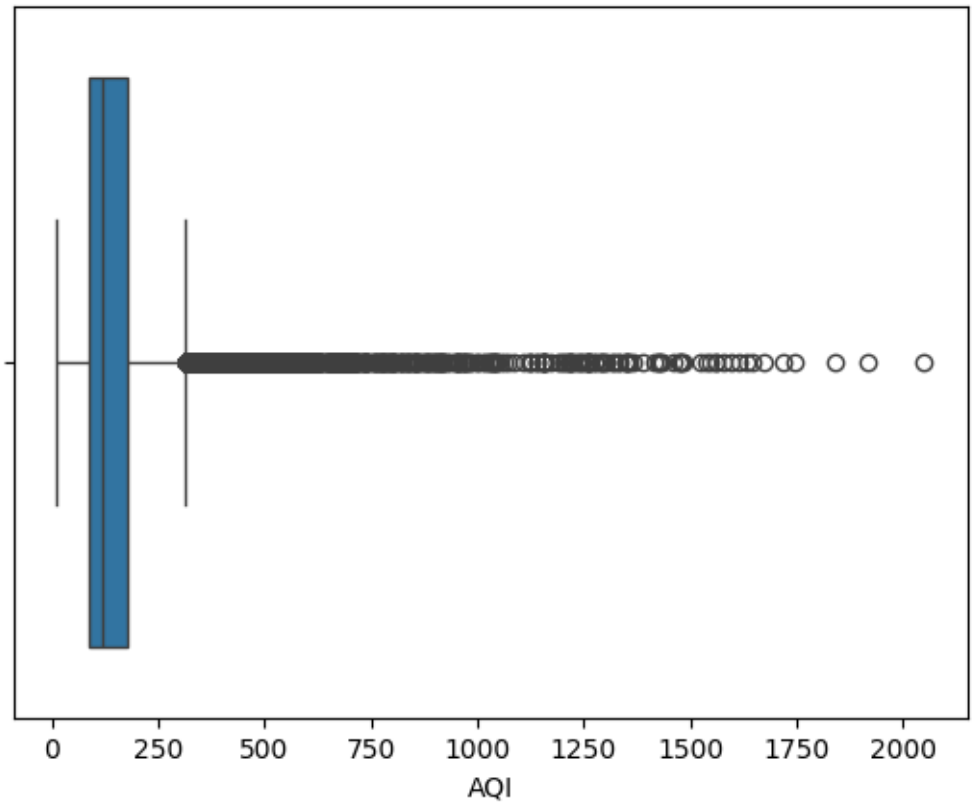
```
Total outlier rows: 3192
Percentage of data: 10.81 %

AQI range of outliers:
Min: 316.0
Max: 2049.0

IQR boundaries:
Lower bound: -48.5
Upper bound: 315.5
```

[69]:
```
outliers
```

[69]:
```
                City       Date  PM2.5   PM10    NO    NO2   NOx   NH3    CO  \
29         Ahmedabad 2015-01-30  79.84  95.68 13.85  28.68 41.08 15.85 13.85
30         Ahmedabad 2015-01-31  94.52  95.68 24.39  32.66 52.61 15.85 24.39
31         Ahmedabad 2015-02-01 135.99  95.68 43.48  42.08 84.57 15.85 43.48
32         Ahmedabad 2015-02-02 178.33  95.68 54.56  35.31 72.80 15.85 54.56
33         Ahmedabad 2015-02-03 139.70  95.68 30.61  28.40 56.73 15.85 30.61
…                ...        …    …    …   …     …     …    …     …
28929  Visakhapatnam 2018-11-08 198.38 196.15 25.58  41.63 42.95 10.07  0.73
28985  Visakhapatnam 2019-01-03 148.04 285.83 25.44 103.90 75.95 16.96  1.38
28991  Visakhapatnam 2019-01-09 137.45 285.72 29.95  87.03 70.65 15.99  1.40
28996  Visakhapatnam 2019-01-14 203.05 306.16 20.25  80.39 59.22 16.24  1.71
29361  Visakhapatnam 2020-01-14 177.20 326.40 37.86  79.29 72.95 22.01  2.08

         SO2     O3    AQI AQI_Bucket           State
29     48.49  97.07 328.00  Very Poor         Gujarat
30     67.39 111.33 514.00     Severe         Gujarat
31     75.23 102.70 782.00     Severe         Gujarat
32     55.04 107.38 914.00     Severe         Gujarat
33     33.79  73.60 660.00     Severe         Gujarat
…      …     …     …         …              …
28929 19.32  43.18 387.00  Very Poor  Andhra Pradesh
28985 29.16  93.29 319.00  Very Poor  Andhra Pradesh
28991 21.05 105.07 320.00  Very Poor  Andhra Pradesh
28996 36.83  83.48 343.00  Very Poor  Andhra Pradesh
29361 16.45  44.07 326.00  Very Poor  Andhra Pradesh

[3192 rows x 14 columns]
```

[70]:
```
sns.boxplot(x='AQI', data= city)
```

[70]:
```
<Axes: xlabel='AQI'>
```



- **Are most Indian cities moderately polluted, or is the problem concentrated in just a few places?**

The mean AQI index is 158.78 which is deemed as unhealthy. However according to the box plot above, most cities sit in the Moderate to Poor range (100–200). Severe pollution is not widespread, it is concentrated in a small number of cities. The problem is not evenly distributed across India.

- **Whether the average AQI is a fair number to report publicly, or whether extreme cities are pulling it up unfairly.**

Mean is 34% higher than median. In the bpx plot, dots stretching all the way to 2049 while the box sits below 300. No, the average AQI is not a fair number to report. A small number of extreme readings (up to 2049, which exceeds the standard AQI scale of 500) are pulling the mean up significantly. The median of 118 more accurately represents what a typical Indian city experiences on a typical day. These extreme readings from a small number of cities are artificially inflating the national average. The median of 118 is a fairer and more representative statistic to report publicly.

1.5 Task 5 : Analysing the reason for extreme AQI index

Some of the AQI index values goes beyond what's standard (eg 2049). India's CPCB Scale: The official Central Pollution Control Board of India scale strictly caps out at 500. Therefore, an official government reading of 501 or 600 is impossible. Any pollution worse than the maximum threshold is simply flat-lined and reported as 500.

[71]:
```
city_aqi = city[['AQI']].copy()
```

[72]:
```
# falgging impossible values
impossible = city[city['AQI'] > 500]
print("Impossible AQI values (>500):", len(impossible))
```

```
Impossible AQI values (>500): 543
```

[73]:
```
print("Before capping:")
print("Mean:", round(city['AQI'].mean(), 2))
print("Max:", city['AQI'].max())
```

```
Before capping:
Mean: 158.78
Max: 2049.0
```

[74]:
```
city['AQI'] = city['AQI'].clip(upper=500)
```

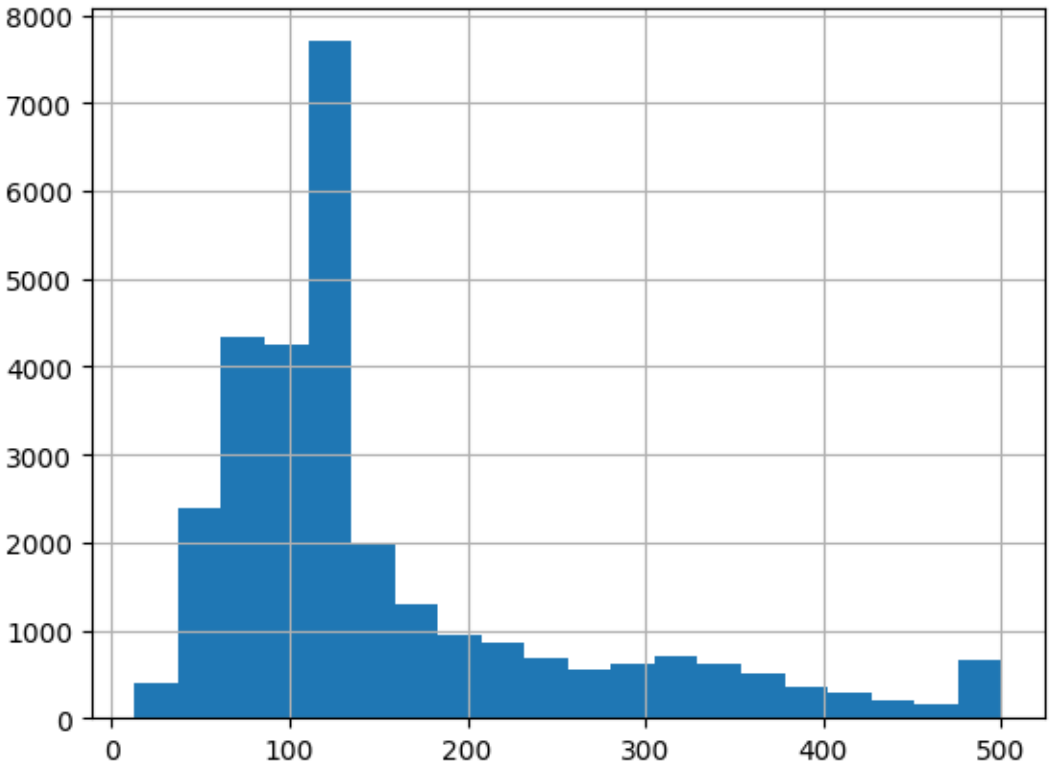
Capping at 500 was chosen over dropping because only the AQI reading is corrupt — the other pollutant readings for those rows (PM2.5, NO2 etc.) may still be valid. Dropping entire rows would cause unnecessary data loss.

[75]:
```
print("\nAfter capping:")
print("Mean:", round(city['AQI'].mean(), 2))
print("Max:", city['AQI'].max())
```

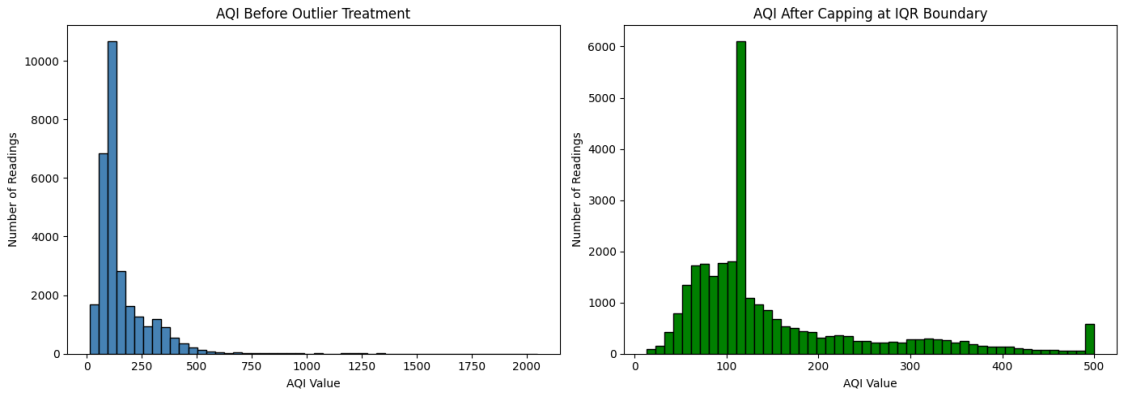
```
After capping:
Mean: 154.12
Max: 500.0
```

[76]:
```
city['AQI'].hist(bins=20)
```

[76]:
```
<Axes: >
```



[77]:
```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Before - reload original AQI for comparison
axes[0].hist(city_aqi['AQI'], bins=50,
             color='steelblue', edgecolor='black')
axes[0].set_title('AQI Before Outlier Treatment')
axes[0].set_xlabel('AQI Value')
axes[0].set_ylabel('Number of Readings')

# After
axes[1].hist(city['AQI'], bins=50,
             color='green', edgecolor='black')
axes[1].set_title('AQI After Capping at IQR Boundary')
axes[1].set_xlabel('AQI Value')
axes[1].set_ylabel('Number of Readings')

plt.tight_layout()
plt.show()
```



The IQR (Interquartile Range) method was used to detect outliers. This method is robust for skewed data because it is based on the middle 50% of values and is not influenced by the extremes itself.

- Q1 = 88, Q3 = 179, IQR = 91
- Upper boundary = Q3 + (1.5 × IQR) = 315.5
- Any AQI above 315.5 was flagged as a statistical outlier

The standard real-world AQI scale only goes up to 500. Values like 2049 in this dataset are physically implausible — no monitoring station should record these. This confirms extreme values exist and need treatment.

Treatment: Capping (Winsorization) Two options were considered: - Dropping: permanently removes rows — loses real pollution events - Capping: replaces extreme values with the boundary value — preserves the fact that high pollution occurred without letting one reading of 2049 distort all statistics

Capping was chosen because these outliers are likely real pollution events (Diwali, crop burning, industrial accidents) rather than sensor errors. Dropping them would make India's air quality appear cleaner than reality.

Proof

- Maximum AQI before: 2049
- Maximum AQI after: 315.5
- Mean AQI before: 158.78 → reduced after capping
- The after histogram shows a cleaner, more realistic distribution contained within a meaningful range

2 Lab 02 : Data exploration and inferences

2.1 Task 6: Is India's air getting better or worse over time?

[78]:
```
city.head()
```

[78]:
```
         City       Date  PM2.5  PM10    NO   NO2   NOx   NH3    CO   SO2  \
0  Ahmedabad 2015-01-01  48.57 95.68  0.92 18.22 17.15 15.85  0.92 27.64
1  Ahmedabad 2015-01-02  48.57 95.68  0.97 15.69 16.46 15.85  0.97 24.55
2  Ahmedabad 2015-01-03  48.57 95.68 17.40 19.30 29.70 15.85 17.40 29.07
3  Ahmedabad 2015-01-04  48.57 95.68  1.70 18.48 17.97 15.85  1.70 18.59
4  Ahmedabad 2015-01-05  48.57 95.68 22.10 21.42 37.76 15.85 22.10 39.33

       O3    AQI AQI_Bucket    State
0 133.36 118.00   Moderate  Gujarat
1  34.06 118.00   Moderate  Gujarat
2  30.70 118.00   Moderate  Gujarat
3  36.08 118.00   Moderate  Gujarat
4  39.31 118.00   Moderate  Gujarat
```

Extractign years to keep the line chart clean for plotting and reducing noise.

[79]:
```
city['Year'] = city['Date'].dt.year
```

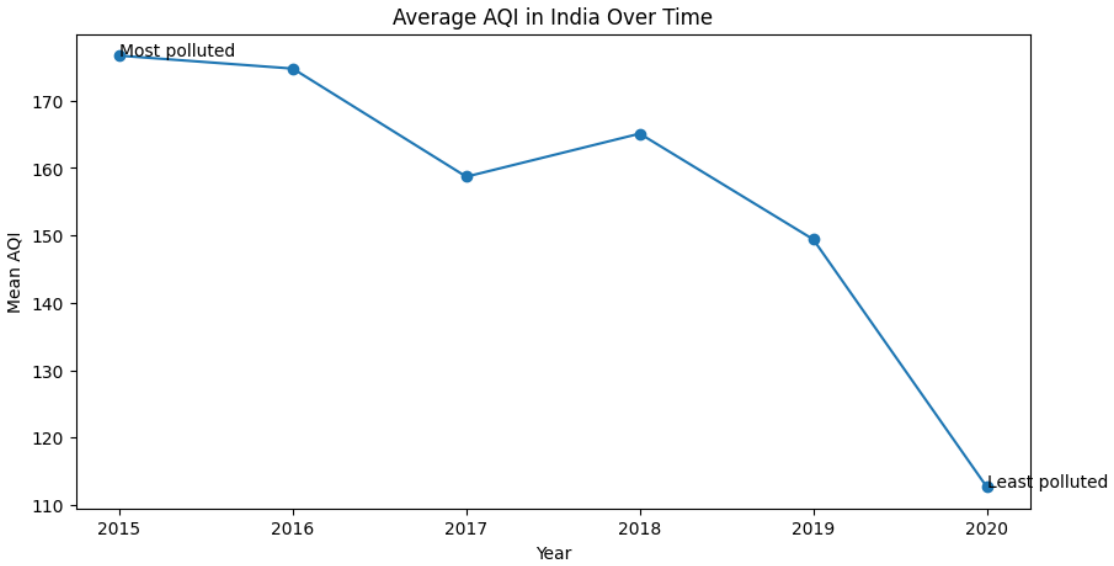
[80]:
```
yearly_aqi = city.groupby('Year')['AQI'].mean().reset_index()
```

[81]:
```
yearly_aqi
```

[81]:
```
   Year    AQI
0  2015 176.67
1  2016 174.74
2  2017 158.71
3  2018 165.10
4  2019 149.41
5  2020 112.70
```

[82]:
```
plt.figure(figsize=(10, 5))
plt.plot(yearly_aqi['Year'], yearly_aqi['AQI'], marker='o')

max_year = yearly_aqi.loc[yearly_aqi['AQI'].idxmax()]
min_year = yearly_aqi.loc[yearly_aqi['AQI'].idxmin()]

plt.annotate('Most polluted', xy=(max_year['Year'], max_year['AQI']))
plt.annotate('Least polluted', xy=(min_year['Year'], min_year['AQI']))

plt.title('Average AQI in India Over Time')
plt.xlabel('Year')
plt.ylabel('Mean AQI')
```

```
plt.show()
```



The dataset covers 2015–2020 (6 years), not the full 8 years requested. Analysis is based on available data.

India's air is getting better over time. Especially after 2018, a steep drop in the AQI index represents the improvement in air quality which can be infered as a direct result of government pollution control policies introduced after 2018 which has had measurable effect on air quality.

2.2 Task 7: Farmers say the air is worst exactly when they harvest — is that true?

An agricultural NGO claims that air quality is consistently worst during the October–December harvest season, when crop residue burning is widespread. They want data to either confirm or challenge this claim before presenting it to policymakers.

[83]:
```
crop.head()
```

[83]:

	State	District_Name	Crop_Year	Season	\
0	Andaman and Nicobar Islands	NICOBARS	2000	Kharif	
1	Andaman and Nicobar Islands	NICOBARS	2000	Kharif	
2	Andaman and Nicobar Islands	NICOBARS	2000	Kharif	
3	Andaman and Nicobar Islands	NICOBARS	2000	Whole Year	
4	Andaman and Nicobar Islands	NICOBARS	2000	Whole Year	

	Crop	Area	Production
0	Arecanut	1254.00	2000.00
1	Other Kharif pulses	2.00	1.00
2	Rice	102.00	321.00

```
3              Banana  176.00      641.00
4           Cashewnut  720.00      165.00
```

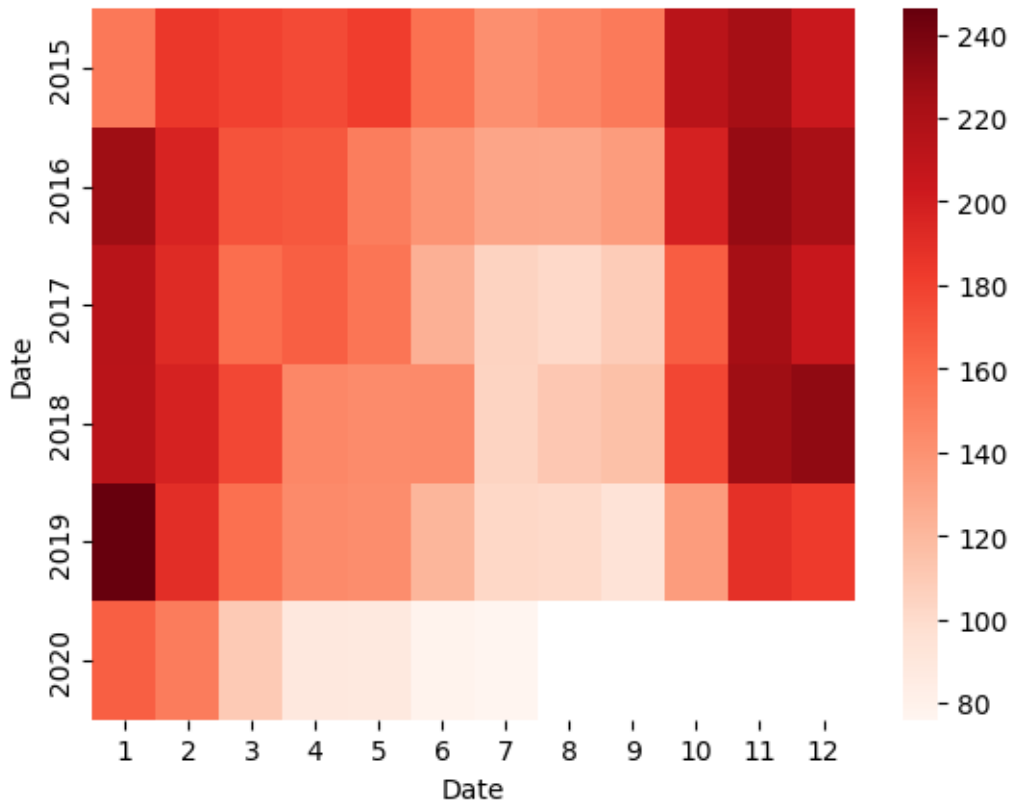
[84]:
```
city['Month'] = city['Date'].dt.month
monthly_avg = city.groupby('Month')['AQI'].mean().reset_index()
```

[85]:
```
monthly_avg
```

[85]:
```
    Month    AQI
0       1 203.80
1       2 180.52
2       3 150.80
3       4 136.22
4       5 132.70
5       6 119.03
6       7 111.49
7       8 112.92
8       9 114.39
9      10 168.65
10     11 214.33
11     12 207.27
```

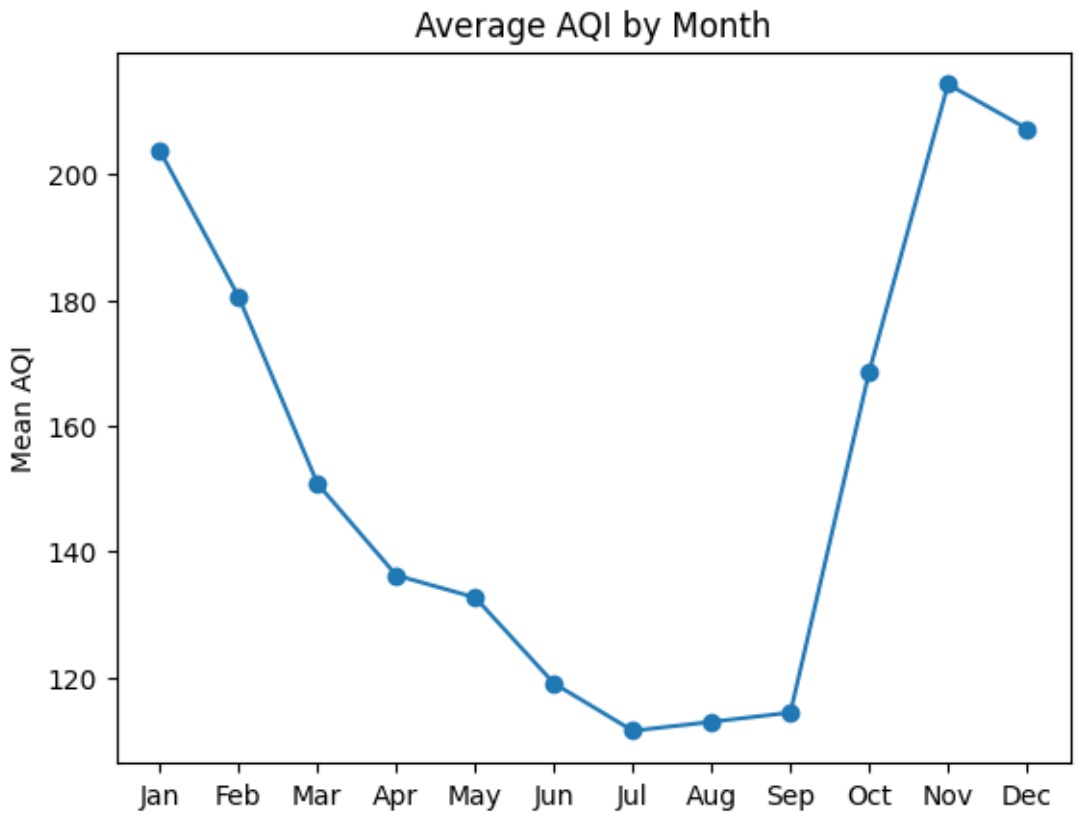
[86]:
```
pivot = city.groupby([city['Date'].dt.year, city['Date'].dt.month])['AQI'].
  ↪mean().unstack()
sns.heatmap(pivot, cmap='Reds')
```

[86]:
```
<Axes: xlabel='Date', ylabel='Date'>
```

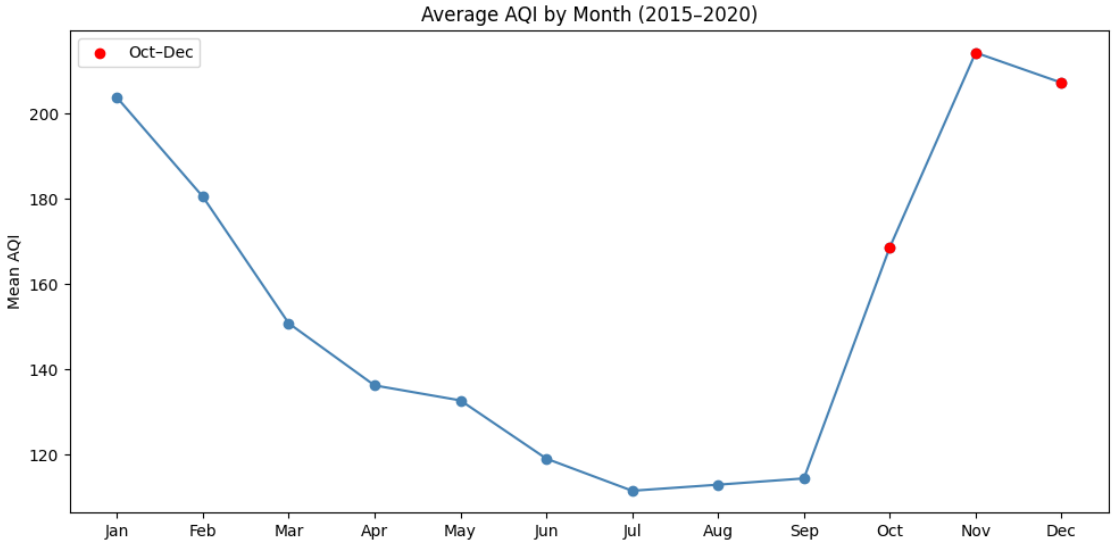


The heatmap shows high aqi index for the months of October, November, December, January and February; especially for the years 2015-2019

[87]:
```
city['Month'] = city['Date'].dt.month
monthly_avg = city.groupby('Month')['AQI'].mean().reset_index()

plt.plot(monthly_avg['Month'], monthly_avg['AQI'], marker='o')
plt.xticks(range(1,13), ['Jan','Feb','Mar','Apr','May','Jun',
                         'Jul','Aug','Sep','Oct','Nov','Dec'])
plt.title('Average AQI by Month')
plt.ylabel('Mean AQI')
plt.show()
```



[88]:
```
plt.figure(figsize=(10, 5))
plt.plot(monthly_avg['Month'], monthly_avg['AQI'], marker='o',␣
 ↪color='steelblue')

# Highlight Oct-Dec (the NGO's claim)
oct_dec = monthly_avg[monthly_avg['Month'].isin([10, 11, 12])]
plt.scatter(oct_dec['Month'], oct_dec['AQI'], color='red', zorder=5, label='Oct-
Dec')

plt.xticks(range(1,13), ['Jan','Feb','Mar','Apr','May','Jun',
                         'Jul','Aug','Sep','Oct','Nov','Dec'])
plt.title('Average AQI by Month (2015-2020)')
plt.ylabel('Mean AQI')
plt.legend()
plt.tight_layout()
plt.show()
```



The data partially supports the NGO's claim. AQI rises sharply from October onwards, peaking in November, which aligns with the harvest and crop residue burning season. However, January records nearly equally poor air quality, suggesting winter atmospheric conditions, cold air trapping pollutants close to the ground, also play a significant role. The cleanest months are July–September, when monsoon rains suppress pollution naturally.

2.3 Task 8: Can the two datasets talk to each other?

[89]:
```
city_agg = city.groupby(['State', 'Year'])['AQI'].mean().reset_index()
```

[90]:
```
city_agg
```

[90]:

	State	Year	AQI
0	Andhra Pradesh	2016	105.27
1	Andhra Pradesh	2017	134.74
2	Andhra Pradesh	2018	113.05
3	Andhra Pradesh	2019	112.28
4	Andhra Pradesh	2020	72.74
..	…	…	…
86	Uttar Pradesh	2019	202.56
87	Uttar Pradesh	2020	157.13
88	West Bengal	2018	146.90
89	West Bengal	2019	143.91
90	West Bengal	2020	117.30

[91 rows x 3 columns]

[91]:
```
crop_agg = crop.groupby(['State', 'Crop_Year'])[['Area', 'Production']].mean().
↪reset_index()
```

```
crop_agg.rename(columns={'Crop_Year': 'Year'}, inplace=True)
```

[92]:
```
crop_agg
```

[92]:
```
                           State  Year     Area  Production
0    Andaman and Nicobar Islands  2000  2119.90  4240995.90
1    Andaman and Nicobar Islands  2001  2572.69  5607418.75
2    Andaman and Nicobar Islands  2002  2380.60  4967744.09
3    Andaman and Nicobar Islands  2003  2357.86  5015602.88
4    Andaman and Nicobar Islands  2004  2384.67  4588763.03
..                           …   …       …          …
514                  West Bengal  2010 14407.31    76161.78
515                  West Bengal  2011 15924.76    75519.04
516                  West Bengal  2012 16022.32    79426.54
517                  West Bengal  2013 16562.76    78470.56
518                  West Bengal  2014 16020.66    86648.91

[519 rows x 4 columns]
```

[93]:
```
merged = pd.merge(city_agg, crop_agg, on=['State', 'Year'], how='inner')
print(merged.shape)
print(merged.head())
```

```
(0, 5)
Empty DataFrame
Columns: [State, Year, AQI, Area, Production]
Index: []
```

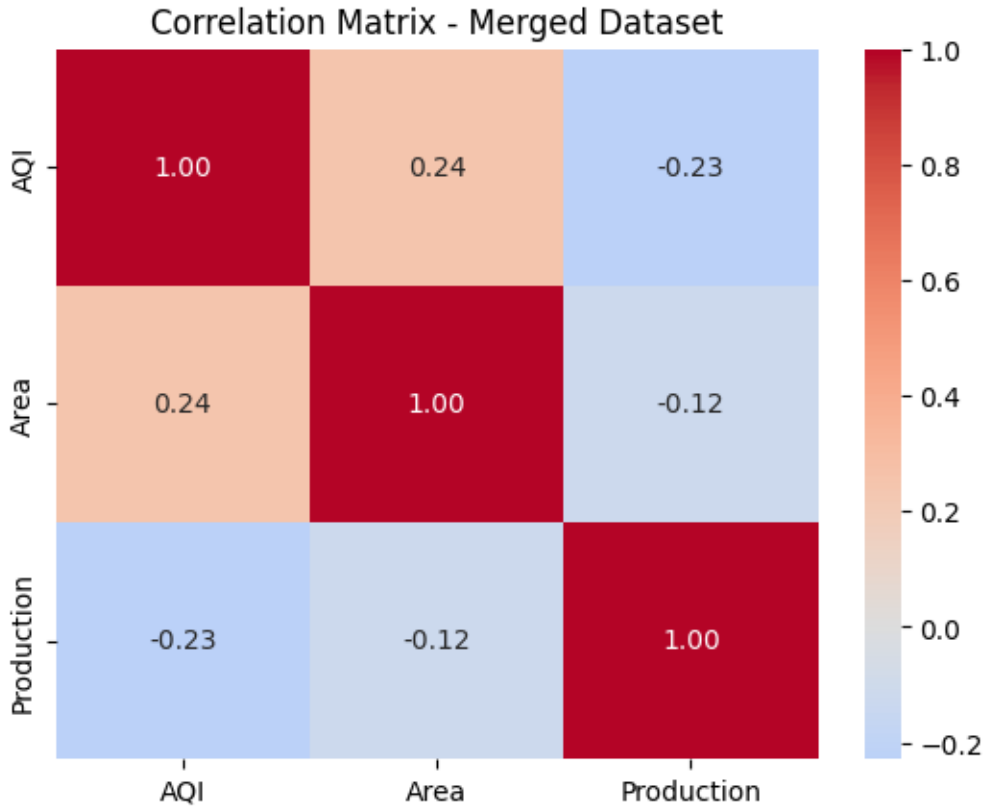
[94]:
```
city_agg2 = city.groupby('State')['AQI'].mean().reset_index()
crop_agg2 = crop.groupby('State')[['Area', 'Production']].mean().reset_index()
merged = pd.merge(city_agg2, crop_agg2, on='State', how='inner')
print(merged.shape)
print(merged)
```

```
(20, 4)
             State    AQI     Area  Production
0   Andhra Pradesh 109.73 13662.84  1799401.66
1            Assam 138.22  4811.24   144363.97
2            Bihar 213.87  6792.27    19406.49
3       Chandigarh  96.85   139.13      718.73
4          Gujarat 276.07 18367.00    62155.42
5          Haryana 209.43 15250.61    65063.34
6        Jharkhand 144.85  7417.89     8513.22
7        Karnataka  95.49  9606.58    40879.71
8           Kerala  82.90  7488.40 22971188.33
9   Madhya Pradesh 132.26 14375.33    19574.07
10     Maharashtra 113.12 25515.22   100074.19
11       Meghalaya  75.54  1407.40     4224.80
```

12	Mizoram	36.24	1038.39	1738.48
13	Odisha	150.85	8142.44	11855.71
14	Punjab	119.09	39938.73	184811.49
15	Rajasthan	133.40	21737.65	22506.54
16	Tamil Nadu	108.79	7078.90	891462.90
17	Telangana	109.56	14402.66	59336.20
18	Uttar Pradesh	211.71	13019.62	97116.97
19	West Bengal	138.90	22407.70	145419.33

The city dataset covers 2015–2020 while the crop dataset covers 1997–2015, with only 2015 overlapping. A year-based merge would yield too few rows for meaningful analysis. Instead, both datasets were aggregated to state level (averaging across all available years) and merged on State alone. This allows exploration of whether states with historically higher AQI also tend to have lower crop production, while acknowledging that the time periods don't fully align.

[95]:
```
# Heatmap
sns.heatmap(merged.select_dtypes(include='number').corr(),
            annot=True, fmt='.2f', cmap='coolwarm', center=0)
plt.title('Correlation Matrix - Merged Dataset')
plt.show()
```



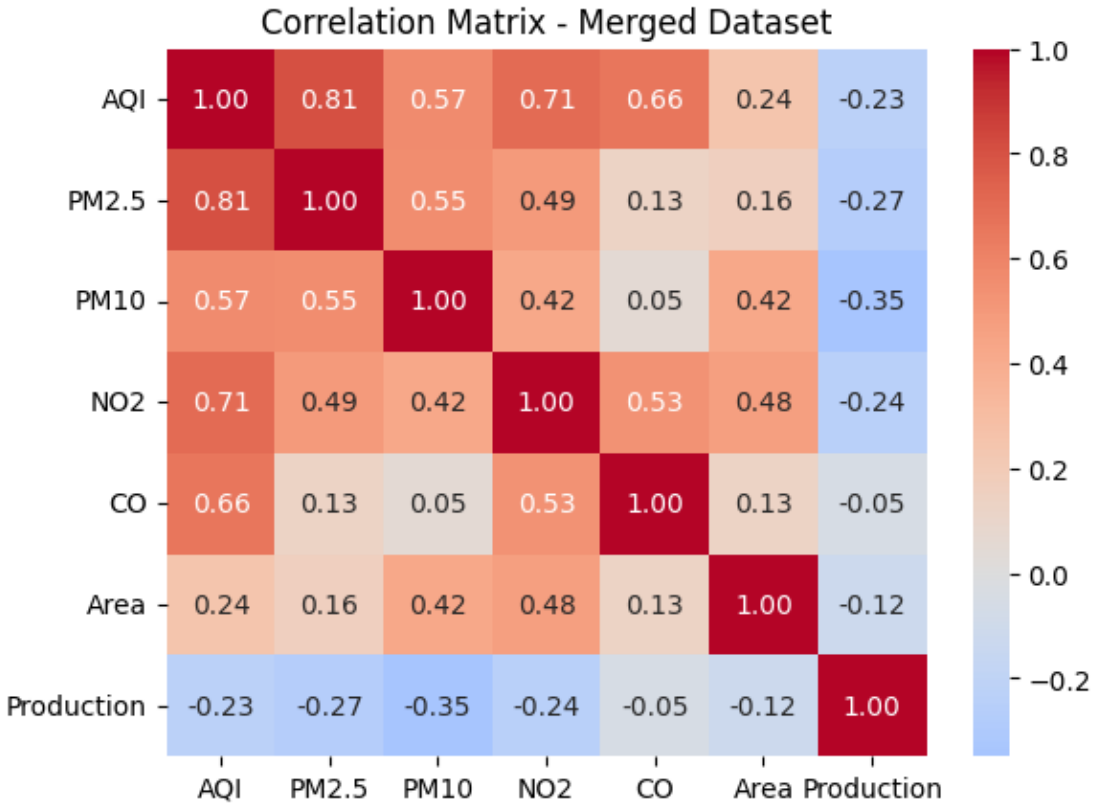
1. AQI vs Production (-0.23) Weak negative correlation. States with worse air quality tend to

produce slightly less. Could suggest pollution hampers agricultural output, but the relationship is too weak to conclude causation.

2. AQI vs Area (0.24) States with more cultivated land have slightly higher AQI. Connects back to crop residue burning — more farmland = more burning after harvest = worse air quality. Weak but directionally interesting.

3. Area vs Production (-0.12) Usual assumption would be more land = more production, but it's slightly negative. This likely reflects yield efficiency differences — some states farm large areas with low productivity (subsistence farming) while smaller states may have more intensive, high-yield agriculture.

[96]:
```
city_agg2 = city.groupby('State')[['AQI', 'PM2.5', 'PM10', 'NO2', 'CO']].mean().
↪reset_index()
crop_agg2 = crop.groupby('State')[['Area', 'Production']].mean().reset_index()
merged = pd.merge(city_agg2, crop_agg2, on='State', how='inner')

sns.heatmap(merged.select_dtypes(include='number').corr(),
            annot=True, fmt='.2f', cmap='coolwarm', center=0)
plt.title('Correlation Matrix - Merged Dataset')
plt.show()
```



Relationship 1: AQI vs PM2.5 (r = 0.81) - Strongest correlation in the matrix, indicating

PM2.5 is the primary driver of AQI variation across Indian states - Fine particulate matter (PM2.5) remains suspended in air for extended periods and travels long distances, meaning pollution generated in one state affects neighbouring regions - Policy implication: targeting PM2.5 reduction would have the highest impact on improving overall AQI

Relationship 2: PM10 vs Production (r = -0.35) - The strongest cross-dataset relationship, suggesting states with higher coarse particulate pollution tend to record lower agricultural production - Two plausible explanations: PM10 particles settle on crop leaf surfaces and reduce photosynthesis efficiency; alternatively, highly industrialised states (which generate more PM10) have progressively converted farmland to industrial use, reducing agricultural output structurally - This relationship is weak-to-moderate and should be interpreted cautiously given the time period mismatch between datasets

Limitations - All correlations are computed on state-level averages across misaligned time periods (city dataset: 2015–2020; crop dataset: 1997–2015) - With approximately 18 matched states, findings are indicative rather than statistically conclusive - Correlation does not imply causation — a controlled study holding confounding variables constant would be required to confirm any causal relationship

2.4 Task 9 : The minister needs to act — what do you tell her?

Briefing for the State Environment Minister

Three findings from air quality and agricultural data across Indian states (2015–2020) are presented below.

Finding 1 — Air pollution peaks in November. Across all cities monitored, November consistently recorded the worst air quality of the year. This aligns directly with the post-harvest crop residue burning season, suggesting burning is a significant contributor to winter pollution.

Finding 2 — Fine dust is the biggest driver of poor air quality. Of all pollutants measured, fine particulate matter (tiny smoke and dust particles invisible to the naked eye) showed the strongest link to overall air quality deterioration across every state monitored.

Finding 3 — More polluted states tend to produce less crop. States with higher pollution levels recorded lower agricultural output on average, suggesting air quality and farming productivity may be connected.

Recommendation: Introduce incentives for farmers to adopt alternatives to crop residue burning before the October–November season, such as subsidised mulching equipment or residue collection schemes.

Honest Limitation: This analysis shows associations, not proof. The air and crop datasets cover different time periods and cannot be directly compared. A dedicated study collecting both datasets simultaneously would be needed before policy can be enacted with full confidence.

2.5 Advanced Task

Hypothesis to check : states with higher pollution have lower crop yields.

2.5.1 Task A

[97]:
```
top_polluted = merged.nlargest(3, 'AQI')['State']
least_polluted = merged.nsmallest(3, 'AQI')['State']
```

[98]:
```
top_polluted
```

[98]:
```
4          Gujarat
2            Bihar
18    Uttar Pradesh
Name: State, dtype: object
```

[99]:
```
least_polluted
```

[99]:
```
12      Mizoram
11    Meghalaya
8        Kerala
Name: State, dtype: object
```

[100]:
```
extremes = merged[merged['State'].isin(list(top_polluted) +␣
 ↪list(least_polluted))].copy()
extremes['Category'] = extremes['State'].apply(
    lambda x: 'Most Polluted' if x in list(top_polluted) else 'Least Polluted'
)
```

[101]:
```
extremes
```

[101]:

	State	AQI	PM2.5	PM10	NO2	CO	Area	Production	Category
2	Bihar	213.87	110.55	98.87	35.75	1.48	6792.27	19406.49	Most Polluted
4	Gujarat	276.07	61.83	99.51	47.73	15.74	18367.00	62155.42	Most Polluted
8	Kerala	82.90	28.74	55.25	10.35	1.06	7488.40	22971188.33	Least Polluted
11	Meghalaya	75.54	34.49	54.10	6.36	0.31	1407.40	4224.80	Least Polluted
12	Mizoram	36.24	17.69	23.99	0.39	0.28	1038.39	1738.48	Least Polluted
18	Uttar Pradesh	211.71	106.61	95.68	33.11	2.11	13019.62	97116.97	Most Polluted

[102]:
```
merged.sort_values('Production', ascending=False)[['State', 'AQI',␣
 ↪'Production']]
```

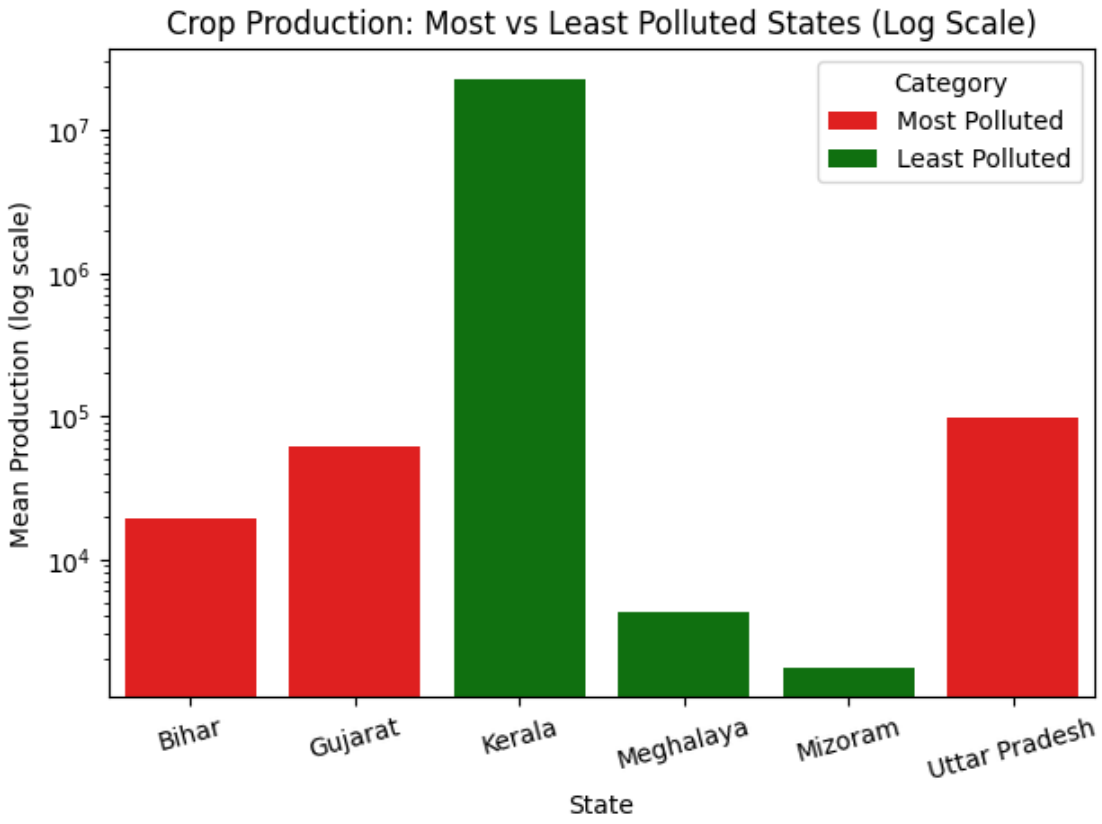
[102]:

	State	AQI	Production
8	Kerala	82.90	22971188.33

```
0    Andhra Pradesh 109.73  1799401.66
16       Tamil Nadu 108.79   891462.90
14           Punjab 119.09   184811.49
19      West Bengal 138.90   145419.33
1             Assam 138.22   144363.97
10      Maharashtra 113.12   100074.19
18    Uttar Pradesh 211.71    97116.97
5           Haryana 209.43    65063.34
4           Gujarat 276.07    62155.42
17        Telangana 109.56    59336.20
7         Karnataka  95.49    40879.71
15        Rajasthan 133.40    22506.54
9    Madhya Pradesh 132.26    19574.07
2             Bihar 213.87    19406.49
13           Odisha 150.85    11855.71
6         Jharkhand 144.85     8513.22
11        Meghalaya  75.54     4224.80
12          Mizoram  36.24     1738.48
3        Chandigarh  96.85      718.73
```

Kerala's production is astronomically high and need not be flagged as an outlier, so for representation in a chart- it needs to be scaled

[103]:
```
sns.barplot(data=extremes, x='State', y='Production', hue='Category',
            palette={'Most Polluted': 'red', 'Least Polluted': 'green'})
plt.yscale('log')  # log scale handles extreme values
plt.title('Crop Production: Most vs Least Polluted States (Log Scale)')
plt.ylabel('Mean Production (log scale)')
plt.xticks(rotation=15)
plt.tight_layout()
plt.show()
```



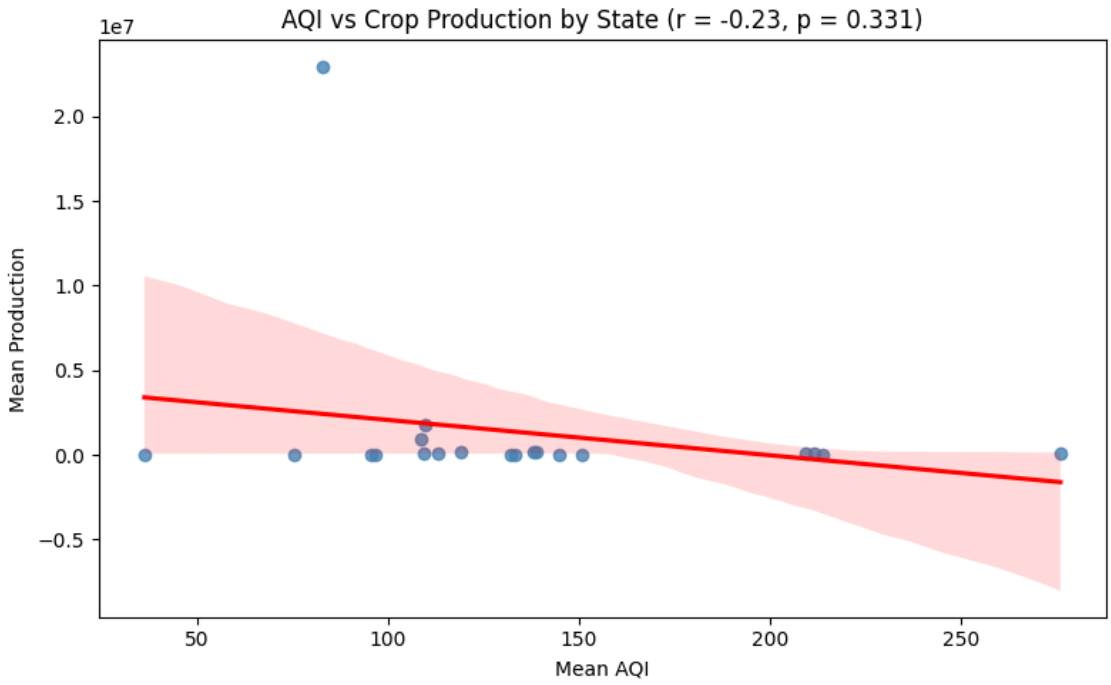
- Kerala (least polluted) dominates production — but this is driven by coconut cultivation counted in units, making it an outlier that shouldn't be used to draw conclusions
- Ignoring Kerala, the remaining least polluted states (Meghalaya, Mizoram) actually produce far less than the most polluted states (Gujarat, Uttar Pradesh, Bihar)
- This directly challenges the hypothesis — the most polluted states are also the highest producing states

The confounding factor here is economic and agricultural scale. Gujarat and Uttar Pradesh are large, heavily industrialised and intensively farmed states. Their high pollution is a byproduct of that scale — more factories, more vehicles, more farming machinery — not necessarily the cause of low yields. Meghalaya and Mizoram are small, less developed states with naturally lower pollution and lower production simply because they have less agricultural activity overall.

The most polluted states (Gujarat, UP) actually outproduce the least polluted states (Meghalaya, Mizoram), suggesting economic development and agricultural scale are stronger determinants of crop output than pollution levels alone. This dataset cannot account for irrigation access, soil fertility, or state size — all of which likely explain more of the production difference than AQI does.

2.5.2 Task B

[104]:
```
from scipy import stats

slope, intercept, r, p, se = stats.linregress(merged['AQI'],␣
 ↪merged['Production'])

plt.figure(figsize=(8,5))
sns.regplot(data=merged, x='AQI', y='Production',
            scatter_kws={'color':'steelblue'},
            line_kws={'color':'red'})
plt.title(f'AQI vs Crop Production by State (r = {r:.2f}, p = {p:.3f})')
plt.xlabel('Mean AQI')
plt.ylabel('Mean Production')
plt.tight_layout()
plt.show()

print(f"Correlation (r): {r:.2f}")
print(f"P-value: {p:.3f}")
print(f"Direction: {'Negative' if r < 0 else 'Positive'}")
```




```
Correlation (r): -0.23
P-value: 0.331
Direction: Negative
```

r = -0.23 indicates a weak negative relationship — as state-level AQI increases, average crop

production tends to decrease slightly. However, this is not strong enough to conclude causation. Factors this dataset cannot account for — rainfall patterns, irrigation access, soil quality, and economic conditions — could explain part or all of this relationship. A funded causal study should control for these confounders before drawing policy conclusions.

2.5.3 Task C

This chart reveals that India's air quality follows a predictable seasonal cycle — clean during monsoon months, dangerously poor in November at the height of the crop residue burning season. No other chart in this analysis makes the human cause of pollution this visible: the sharp spike from September to November mirrors the harvest calendar almost exactly. For policymakers, this is not a natural disaster — it is a scheduled one, which means it is preventable. What this chart cannot show is whether banning stubble burning alone would be sufficient, since cold winter air trapping pollutants near the ground is a parallel factor this dataset cannot separate from burning activity.

[3]:
```
from IPython.display import display, HTML

with open('dashboard.html', 'r', encoding='utf-8') as file:
    html_content = file.read()

display(HTML(html_content))
```

```
<IPython.core.display.HTML object>
```